NACCA COURSES • MATHEMATICS

MATHS QUEST

BASIC 6

Explore 48 adventures in number, algebra, geometry, measurement and data.

48
LESSONS

44
CURRICULUM INDICATORS

720
ORIGINAL PRACTICE ITEMS

PRINTABLE LEARNER'S GUIDE · 2026 LEARNING EDITION

Your Maths Quest Starts Here

This book helps you think like a mathematician. Explore an example, try five guided Play tasks, then solve ten different Challenge problems.

LOOK & EXPLORE

Read the lesson goal, investigate its diagram and follow a worked example.

PLAY · 5 QUESTIONS

Use what you have just learned. Check your methods with a friend or adult.

CHALLENGE · 10 QUESTIONS

Try new numbers and fresh situations. Show your method, not only your answer.

REFLECT

Explain why your method works. Use the answer keys for self-checking after you try.

SAFETY AND SMART PRACTICE

Use clean counters, paper, pencils and a ruler. Ask an adult before going outside to measure, observe traffic, or handle tools. All activities can be done safely with drawings.

Find Your Next Adventure

01	Representing numbers to one million	Page 6
02	Reading numbers to one billion	Page 9
03	Finding neighbours on large number charts	Page 12
04	Comparing and ordering six-digit numbers	Page 15
05	Rounding large whole numbers	Page 18
06	Skip counting in thousands	Page 21
07	Recognising Roman numerals to C	Page 24
08	Converting Roman and Arabic numerals	Page 27
09	Highest common factors by prime factors	Page 30
10	Least common multiples by prime factors	Page 33
11	Mental maths and number properties	Page 36
12	Efficient mental multiplication	Page 39
13	Multiplication with multi-digit factors	Page 42
14	Division fact families	Page 45
15	Three-digit division	Page 48
16	Multi-step problems with four operations	Page 51
17	Locating and comparing integers	Page 54
18	Adding and subtracting integers	Page 57
19	Multiplication with positive and negative integers	Page 60
20	Comparing fractions, decimals and percentages	Page 63
21	Adding and subtracting unlike fractions	Page 66

Find Your Next Adventure

22	Multiplying fractions by whole numbers	Page 69
23	Representing ratios with objects	Page 72
24	Making and comparing equivalent ratios	Page 75
25	Recognising proportional relationships	Page 78
26	Solving proportional reasoning problems	Page 81
27	Visual number patterns and predictions	Page 84
28	Writing algebraic pattern rules	Page 87
29	Describing number tables with expressions	Page 90
30	Writing algebraic expressions	Page 93
31	Combining like terms	Page 96
32	Substituting values in expressions	Page 99
33	Solving one-step equations	Page 102
34	Creating real stories for equations	Page 105
35	Recognising rectangular and triangular prisms	Page 108
36	Nets of prisms	Page 111
37	Cardinal directions and routes	Page 114
38	Reflecting shapes across a mirror line	Page 117
39	Translating shapes across a grid	Page 120
40	Labelling and reading line graphs	Page 123
41	Choosing when a line graph is suitable	Page 126
42	Plotting and interpreting line graphs	Page 129

Find Your Next Adventure

43	Choosing how to collect data	Page 132
44	Designing useful survey questions	Page 135
45	Organising survey results	Page 138
46	Listing outcomes of chance experiments	Page 141
47	Predicting chances and likelihood	Page 144
48	Comparing experimental and theoretical probability	Page 147

Representing numbers to one million

LEARNING GOAL Representing numbers to one million

MATHS MODEL • PLACE



Explore each place • Compare • Explain

LEARN THE IDEA

A digit has a different value in each place: hundreds, thousands, ten thousands, hundred thousands and millions. multiply the digit by its place value.

WORKED EXAMPLE 1

In 524,668, what value does the digit in the 100s column represent?

Answer: 600

6 × 100 = 600.

WORKED EXAMPLE 2

In 415,318, what value does the digit in the 10s column represent?

Answer: 10

1 × 10 = 10.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Representing numbers to one million

PLAY · LEARN THE STEPS

1. **In 174,769, what value does the digit in the 100,000s column represent?**
   A) 10,000 B) 1,000,000 C) 100,000 D) 1
2. **In 583,817, what value does the digit in the 10,000s column represent?**
   A) 8 B) 80,000 C) 800,000 D) 8,000
3. **In 529,071, what value does the digit in the 1,000s column represent?**
   A) 90,000 B) 900 C) 9 D) 9,000
4. **In 521,922, what value does the digit in the 1,000s column represent?**
   A) 1 B) 10,000 C) 100 D) 1,000
5. **In 134,524, what value does the digit in the 100,000s column represent?**
   A) 1 B) 10,000 C) 100,000 D) 1,000,000

CHALLENGE · DIFFERENT EXAMPLES

1. **A pupil at a market stall removes the contribution of the 1,000s digit from 938,988. What number remains?**
   A) 938,988 B) 931,988 C) 930,988 D) 938,980
2. **A record during a library project shows 780,719. What is the value of the digit in the 100s column?**
   A) 7,000 B) 7 C) 700 D) 70
3. **A six-digit number on a football field has 789,596 on a display. Which digit represents the 1,000s place?**
   A) 4 B) 2 C) 9 D) 0
4. **A pupil in a recycling club removes the contribution of the 10s digit from 324,127. What number remains?**
   A) 324,125 B) 324,107 C) 324,117 D) 324,127
5. **A record at a science exhibition shows 865,213. What is the value of the digit in the 10,000s column?**
   A) 6 B) 600,000 C) 60,000 D) 6,000

Representing numbers to one million

CHALLENGE · APPLY WHAT YOU KNOW

6 **A record during a reading competition shows 301,592. What is the value of the digit in the 1,000s column?**

A) 100 B) 1 C) 1,000 D) 10,000

7 **A six-digit number for a school fundraiser has 454,493 on a display. Which digit represents the 1,000s place?**

A) 7 B) 4 C) 9 D) 5

8 **A pupil at a farm removes the contribution of the 100s digit from 467,965. What number remains?**

A) 467,965 B) 467,065 C) 467,165 D) 467,956

9 **A record during a sports day shows 695,663. What is the value of the digit in the 10,000s column?**

A) 9,000 B) 900,000 C) 9 D) 90,000

10 **A six-digit number in a craft workshop has 490,418 on a display. Which digit represents the 100,000s place?**

A) 5 B) 7 C) 9 D) 4

ANSWER CHECK · LOOK ONLY AFTER TRYING

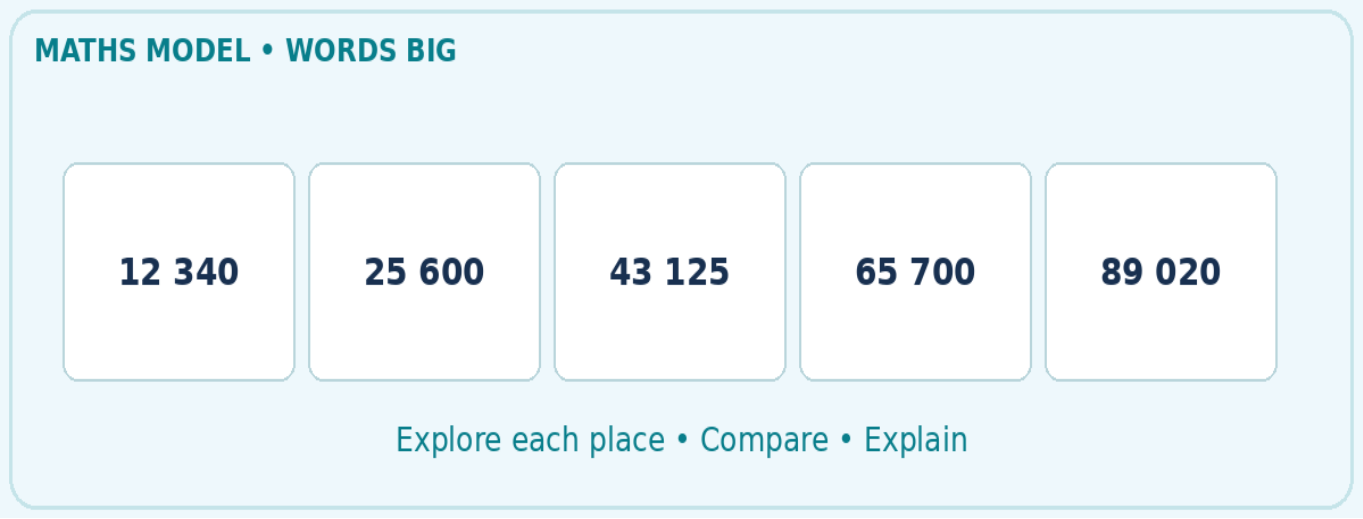
1. 100,000	2. 80,000	3. 9,000
4. 1,000	5. 100,000	6. 930,988
7. 700	8. 9	9. 324,107
10. 60,000	11. 1,000	12. 4
13. 467,065	14. 90,000	15. 4

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Reading numbers to one billion

LEARNING GOAL Reading numbers to one billion



LEARN THE IDEA

Read digits from left to right in groups of three: millions, thousands and ones. a thousand millions is one billion.

WORKED EXAMPLE 1

Which words name 46,772,000?

Answer: forty-six million seven hundred and seventy-two thousand

Read millions, then thousands, then units: forty-six million seven hundred and seventy-two thousand.

WORKED EXAMPLE 2

Write “one hundred and twenty-five million four hundred and fifty-six thousand” in figures.

Answer: 125,456,000

The millions group and thousands group give 125,456,000.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Reading numbers to one billion

PLAY · LEARN THE STEPS

1. **Which words name 417,840,000?**
   A) four hundred and eighteen million eight hundred and forty thousand B) four hundred and seventeen million eight hundred and forty thousand C) four hundred and sixteen million eight hundred and forty thousand D) four hundred and seventeen million eight hundred and forty-one thousand

2. **Write “forty-seven million two hundred and fifty-one thousand” in figures.**
   A) 48,251,000 B) 47,251,000 C) 46,251,000 D) 47,252,000

3. **Which words name 372,428,000?**
   A) three hundred and seventy-two million four hundred and twenty-eight thousand B) three hundred and seventy-two million four hundred and twenty-nine thousand C) three hundred and seventy-three million four …

4. **Write “two hundred and forty-seven million nine hundred and eight thousand” in figures.**
   A) 247,909,000 B) 248,908,000 C) 246,908,000 D) 247,908,000

5. **Which words name 424,540,000?**
   A) four hundred and twenty-five million five hundred and forty thousand B) four hundred and twenty-four million five hundred and forty-one thousand C) four hundred and twenty-three million five hundred and forty thousand D) four hundred and twenty-four million five hundred and forty thousand

CHALLENGE · DIFFERENT EXAMPLES

1. **A records office at a science exhibition writes “six hundred and six million sixty-four thousand”. Select the matching numeral.**
   A) 606,065,000 B) 607,064,000 C) 606,064,000 D) 605,064,000

2. **A public report during a reading competition shows 183,646,000. Choose its number name.**
   A) one hundred and eighty-four million six hundred and forty-six thousand B) one hundred and eighty-three million six hundred and forty-six thousand C) one hundred and eighty-three million six hundred and forty-seven thousand D) one hundred and eighty-two million six hundred and forty-six thousand

3. **A records office for a school fundraiser writes “two hundred and forty-eight million three hundred and sixty-six thousand”. Select the matching numeral.**
   A) 249,366,000 B) 247,366,000 C) 248,366,000 D) 248,367,000

4. **A public report at a farm shows 979,183,000. Choose its number name.**
   A) nine hundred and seventy-nine million one hundred and eighty-four thousand B) nine hundred and seventy-nine million one hundred and eighty-three thousand C) nine hundred and seventy-eight million one hundred and eighty-three thousand D) nine hundred and eighty million one hundred and eighty-three thousand

5. **A records office during a sports day writes “three hundred and twenty-six million one hundred and fifty-three thousand”. Select the matching numeral.**
   A) 326,153,000 B) 327,153,000 C) 325,153,000 D) 326,154,000

Reading numbers to one billion

CHALLENGE · APPLY WHAT YOU KNOW

6 **A public report in a craft workshop shows 740,742,000. Choose its number name.**

A) seven hundred and thirty-nine million seven hundred and forty-two thousand B) seven hundred and forty million seven hundred and forty-three thousand C) seven hundred and forty-one million seven hundred and forty-two thousand D) seven hundred and forty million seven hundred and forty-two thousand

7 **A records office at a school fair writes “nine hundred and thirty-six million seven hundred and thirty-three thousand”. Select the matching numeral.**

A) 936,734,000 B) 936,733,000 C) 937,733,000 D) 935,733,000

8 **A public report in the class garden shows 771,272,000. Choose its number name.**

A) seven hundred and seventy-two million two hundred and seventy-two thousand B) seven hundred and seventy-one million two hundred and seventy-two thousand C) seven hundred and seventy million two hundred and seventy-two thousand D) seven hundred and seventy-one million two hundred and seventy-three thousand

9 **A records office at a market stall writes “one hundred and fifty-nine million two hundred and fifty-eight thousand”. Select the matching numeral.**

A) 159,258,000 B) 158,258,000 C) 160,258,000 D) 159,259,000

10 **A public report during a library project shows 626,774,000. Choose its number name.**

A) six hundred and twenty-six million seven hundred and seventy-four thousand B) six hundred and twenty-seven million seven hundred and seventy-four thousand C) six hundred and twenty-five million seven hundred and seventy-four thousand D) six hundred and twenty-six million seven hundred and seventy-five thousand

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. four hundred and seventeen …
2. 47,251,000
3. three hundred and seventy-two …
4. 247,908,000
5. four hundred and twenty-four …
6. 606,064,000
7. one hundred and eighty-three …
8. 248,366,000
9. nine hundred and seventy-nine …
10. 326,153,000
11. seven hundred and forty million …
12. 936,733,000
13. seven hundred and seventy-one …
14. 159,258,000
15. six hundred and twenty-six …

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Finding neighbours on large number charts

LEARNING GOAL Finding neighbours on large number charts

MATHS MODEL • CHART

25 000	25 500	26 000	26 500	27 000

Explore each place • Compare • Explain

LEARN THE IDEA

On a number chart, check the size of a jump. moving right or forward adds; moving left or backward subtracts.

WORKED EXAMPLE 1

What is 10 less than 58,250?

Answer: 58,240

58,250 − 10 = 58,240.

WORKED EXAMPLE 2

What is 1 more than 11,733?

Answer: 11,734

11,733 + 1 = 11,734.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Finding neighbours on large number charts

PLAY · LEARN THE STEPS

1. **What is 1 more than 9,955?**
   A) 9,954 B) 9,965 C) 9,957 D) 9,956
2. **What is 100 more than 43,034?**
   A) 43,234 B) 44,034 C) 43,134 D) 42,934
3. **What is 10 more than 26,288?**
   A) 26,388 B) 26,298 C) 26,308 D) 26,278
4. **What is 100 less than 41,658?**
   A) 40,658 B) 41,758 C) 41,558 D) 41,458
5. **What is 1 less than 43,380?**
   A) 43,378 B) 43,370 C) 43,379 D) 43,381

CHALLENGE · DIFFERENT EXAMPLES

1. **A chart on a football field lists steps of 1. The highlighted box is 49,827; what number is immediately after it?**
   A) 49,829 B) 49,837 C) 49,826 D) 49,828
2. **A chart in a recycling club lists steps of 10. The highlighted box is 25,776; what number is immediately after it?**
   A) 25,876 B) 25,786 C) 25,766 D) 25,796
3. **A chart at a science exhibition lists steps of 100. The highlighted box is 34,209; what number is immediately after it?**
   A) 34,409 B) 34,309 C) 35,209 D) 34,109
4. **A chart during a reading competition lists steps of 100. The highlighted box is 18,807; what number is immediately before it?**
   A) 18,607 B) 18,907 C) 17,807 D) 18,707
5. **A chart for a school fundraiser lists steps of 100. The highlighted box is 72,651; what number is immediately after it?**
   A) 72,851 B) 72,551 C) 72,751 D) 73,651

Finding neighbours on large number charts

CHALLENGE · APPLY WHAT YOU KNOW

6 **A chart at a farm lists steps of 1000. The highlighted box is 54,645; what number is immediately after it?**

A) 64,645 B) 55,645 C) 56,645 D) 53,645

7 **A chart during a sports day lists steps of 1000. The highlighted box is 62,413; what number is immediately before it?**

A) 52,413 B) 63,413 C) 60,413 D) 61,413

8 **A chart in a craft workshop lists steps of 1. The highlighted box is 37,378; what number is immediately before it?**

A) 37,376 B) 37,368 C) 37,379 D) 37,377

9 **A chart at a school fair lists steps of 1000. The highlighted box is 44,262; what number is immediately after it?**

A) 46,262 B) 45,262 C) 43,262 D) 54,262

10 **A chart in the class garden lists steps of 1000. The highlighted box is 26,326; what number is immediately before it?**

A) 27,326 B) 16,326 C) 24,326 D) 25,326

ANSWER CHECK · LOOK ONLY AFTER TRYING

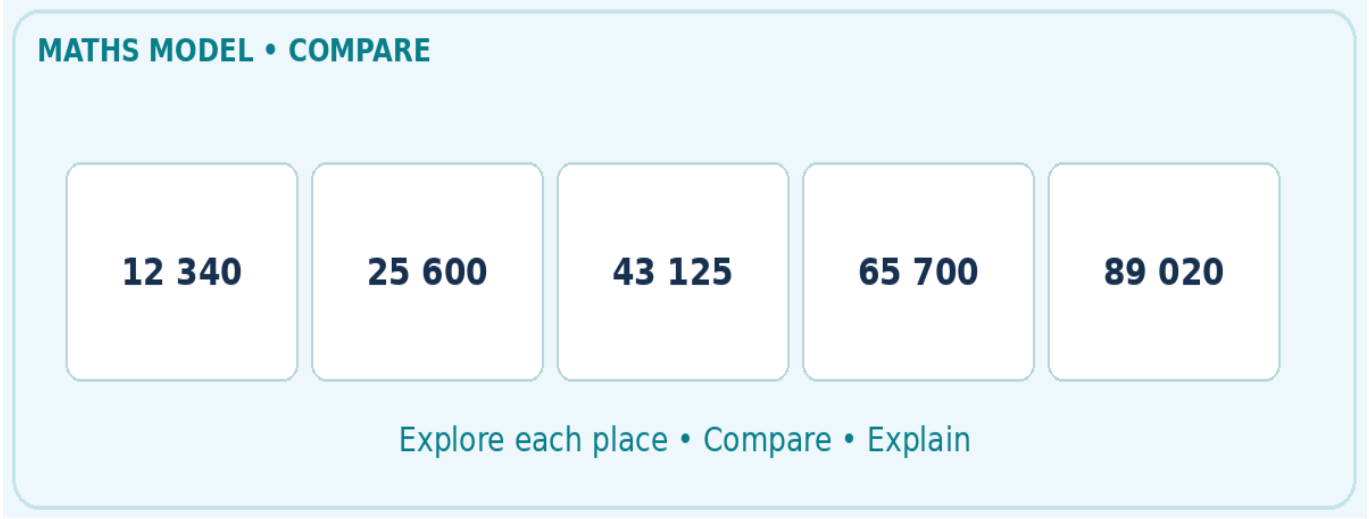
1. 9,956	2. 43,134	3. 26,298
4. 41,558	5. 43,379	6. 49,828
7. 25,786	8. 34,309	9. 18,707
10. 72,751	11. 55,645	12. 61,413
13. 37,377	14. 45,262	15. 25,326

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Comparing and ordering six-digit numbers

LEARNING GOAL Comparing and ordering six-digit numbers



LEARN THE IDEA

Compare whole numbers from the highest place value. use < for less than, > for greater than and = for equality.

WORKED EXAMPLE 1

Choose the larger number: 89,830 or 93,149.

Answer: 93,149

Compare the digits from left to right; 93,149 is larger.

WORKED EXAMPLE 2

Which sign makes 90,403 __ 93,995 true?

Answer: <

90,403 is less than 93,995.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Comparing and ordering six-digit numbers

PLAY · LEARN THE STEPS

1. **Choose the larger number: 19,692 or 17,923.**
   A) 17,823 B) 19,692 C) 20,692 D) 17,923
2. **Which sign makes 22,968 __ 21,031 true?**
   A) none of these B) = C) > D) <
3. **Which sign makes 46,676 __ 47,979 true?**
   A) = B) > C) < D) none of these
4. **Choose the larger number: 76,207 or 76,778.**
   A) 76,207 B) 76,778 C) 76,107 D) 77,778
5. **Which sign makes 45,355 __ 48,890 true?**
   A) none of these B) = C) > D) <

CHALLENGE · DIFFERENT EXAMPLES

1. **Fill the comparison on the scoreboard: 17,566 __ 19,233.**
   A) < B) = C) none of these D) >
2. **Two totals at a science exhibition are 35,142 and 33,339. Which is greater?**
   A) 33,239 B) 35,142 C) 36,142 D) 33,339
3. **Fill the comparison on the scoreboard: 20,674 __ 20,701.**
   A) none of these B) = C) < D) >
4. **Fill the comparison on the scoreboard: 17,028 __ 17,293.**
   A) > B) < C) none of these D) =
5. **Two totals at a farm are 64,309 and 63,708. Which is greater?**
   A) 63,708 B) 64,309 C) 65,309 D) 63,608

Comparing and ordering six-digit numbers

CHALLENGE · APPLY WHAT YOU KNOW

6. **Two totals during a sports day are 11,229 and 13,994. Which is greater?**
A) 13,994 B) 11,229 C) 11,129 D) 14,994

7. **Fill the comparison on the scoreboard: 29,455 __ 30,234.**
A) > B) none of these C) < D) =

8. **Fill the comparison on the scoreboard: 47,479 __ 49,605.**
A) > B) none of these C) = D) <

9. **Two totals in the class garden are 54,761 and 56,928. Which is greater?**
A) 56,928 B) 54,761 C) 57,928 D) 54,661

10. **Fill the comparison on the scoreboard: 39,742 __ 37,715.**
A) > B) = C) < D) none of these

ANSWER CHECK · LOOK ONLY AFTER TRYING

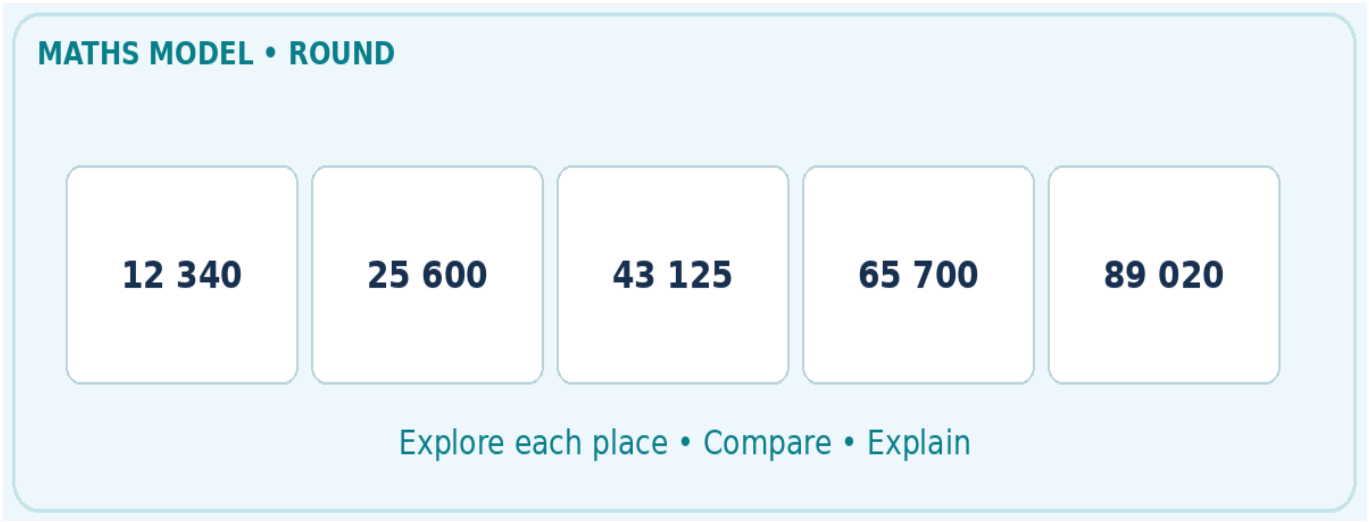
1. 19,692	2. >	3. <
4. 76,778	5. <	6. <
7. 35,142	8. <	9. <
10. 64,309	11. 13,994	12. <
13. <	14. 56,928	15. >

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Rounding large whole numbers

LEARNING GOAL Rounding large whole numbers



LEARN THE IDEA

To round, find the target place and inspect the digit immediately to its right. five or more rounds up.

WORKED EXAMPLE 1

Round 55,994 to the nearest 100.

Answer: 56,000

Check the next digit and round 55,994 to 56,000.

WORKED EXAMPLE 2

Round 40,606 to the nearest 10.

Answer: 40,610

Check the next digit and round 40,606 to 40,610.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Rounding large whole numbers

PLAY · LEARN THE STEPS

1. **Round 87,600 to the nearest 10.**
   A) 87,610 B) 87,600 C) 87601 D) 87,590
2. **Round 17,674 to the nearest 10,000.**
   A) 20,000 B) 10,000 C) 30,000 D) 17,674
3. **Round 90,034 to the nearest 10,000.**
   A) 100,000 B) 90,034 C) 90,000 D) 80,000
4. **Round 16,538 to the nearest 10.**
   A) 16,538 B) 16,530 C) 16,540 D) 16,550
5. **Round 33,987 to the nearest 100.**
   A) 34,100 B) 33,987 C) 34,000 D) 33,900

CHALLENGE · DIFFERENT EXAMPLES

1. **A crowd estimate at a science exhibition is based on 54,130 visitors. To the nearest 1,000, what should be reported?**
   A) 55,000 B) 54,130 C) 53,000 D) 54,000
2. **A crowd estimate during a reading competition is based on 85,011 visitors. To the nearest 1,000, what should be reported?**
   A) 85,000 B) 86,000 C) 85,011 D) 84,000
3. **A crowd estimate for a school fundraiser is based on 16,658 visitors. To the nearest 100, what should be reported?**
   A) 16,658 B) 16,800 C) 16,700 D) 16,600
4. **A crowd estimate at a farm is based on 76,562 visitors. To the nearest 100, what should be reported?**
   A) 76,500 B) 76,562 C) 76,700 D) 76,600
5. **A crowd estimate during a sports day is based on 17,176 visitors. To the nearest 100, what should be reported?**
   A) 17,100 B) 17,200 C) 17,300 D) 17,176

Rounding large whole numbers

CHALLENGE · APPLY WHAT YOU KNOW

6 **A crowd estimate in a craft workshop is based on 87,739 visitors. To the nearest 10, what should be reported?**

A) 87,750 B) 87,739 C) 87,730 D) 87,740

7 **A crowd estimate at a school fair is based on 10,906 visitors. To the nearest 10,000, what should be reported?**

A) 10,906 B) 10,000 C) 0 D) 20,000

8 **A crowd estimate in the class garden is based on 42,948 visitors. To the nearest 10,000, what should be reported?**

A) 30,000 B) 42,948 C) 40,000 D) 50,000

9 **A crowd estimate at a market stall is based on 53,979 visitors. To the nearest 1,000, what should be reported?**

A) 53,000 B) 55,000 C) 53,979 D) 54,000

10 **A crowd estimate during a library project is based on 70,269 visitors. To the nearest 10, what should be reported?**

A) 70,270 B) 70,269 C) 70,260 D) 70,280

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 87,600	2. 20,000	3. 90,000
4. 16,540	5. 34,000	6. 54,000
7. 85,000	8. 16,700	9. 76,600
10. 17,200	11. 87,740	12. 10,000
13. 40,000	14. 54,000	15. 70,270

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Skip counting in thousands

LEARNING GOAL Skip counting in thousands

MATHS MODEL • SKIP



Explore each place • Compare • Explain

LEARN THE IDEA

A skip-counting pattern adds or subtracts the same step at every landing. check both forward and backward travel.

WORKED EXAMPLE 1

Count 4 jumps forward in 2,000s from 42,000.

Answer: 50,000

42,000 + 4 × 2,000 = 50,000.

WORKED EXAMPLE 2

Count 3 jumps forward in 2,000s from 34,000.

Answer: 40,000

34,000 + 3 × 2,000 = 40,000.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Skip counting in thousands

PLAY · LEARN THE STEPS

1. **Count 5 jumps back in 2,000s from 50,000.**
   A) 40,000 B) 60,000 C) 42,000 D) 38,000
2. **Count 6 jumps forward in 1,000s from 34,000.**
   A) 41,000 B) 28,000 C) 40,000 D) 39,000
3. **Count 6 jumps forward in 2,000s from 36,000.**
   A) 46,000 B) 50,000 C) 24,000 D) 48,000
4. **Count 4 jumps forward in 1,000s from 12,000.**
   A) 8,000 B) 15,000 C) 17,000 D) 16,000
5. **Count 5 jumps back in 2,500s from 47,500.**
   A) 35,000 B) 60,000 C) 32,500 D) 37,500

CHALLENGE · DIFFERENT EXAMPLES

1. **A route during a reading competition moves 4 jumps of 1,000 backward from 19,000. Where does it finish?**
   A) 14,000 B) 16,000 C) 15,000 D) 23,000
2. **A route for a school fundraiser moves 5 jumps of 2,000 backward from 66,000. Where does it finish?**
   A) 76,000 B) 58,000 C) 54,000 D) 56,000
3. **A route at a farm moves 6 jumps of 1,000 forward from 23,000. Where does it finish?**
   A) 29,000 B) 28,000 C) 17,000 D) 30,000
4. **A route during a sports day moves 2 jumps of 1,000 forward from 18,000. Where does it finish?**
   A) 16,000 B) 19,000 C) 21,000 D) 20,000
5. **A route in a craft workshop moves 5 jumps of 2,500 forward from 30,000. Where does it finish?**
   A) 42,500 B) 17,500 C) 40,000 D) 45,000

Skip counting in thousands

CHALLENGE · APPLY WHAT YOU KNOW

6 **A route at a school fair moves 6 jumps of 2,000 backward from 32,000. Where does it finish?**

A) 22,000 B) 18,000 C) 20,000 D) 44,000

7 **A route in the class garden moves 3 jumps of 500 backward from 13,000. Where does it finish?**

A) 12,000 B) 11,500 C) 11,000 D) 14,500

8 **A route at a market stall moves 3 jumps of 1,000 forward from 17,000. Where does it finish?**

A) 21,000 B) 19,000 C) 20,000 D) 14,000

9 **A route during a library project moves 6 jumps of 500 backward from 9,500. Where does it finish?**

A) 12,500 B) 6,500 C) 6,000 D) 7,000

10 **A route on a football field moves 2 jumps of 2,500 forward from 37,500. Where does it finish?**

A) 40,000 B) 45,000 C) 42,500 D) 32,500

ANSWER CHECK · LOOK ONLY AFTER TRYING

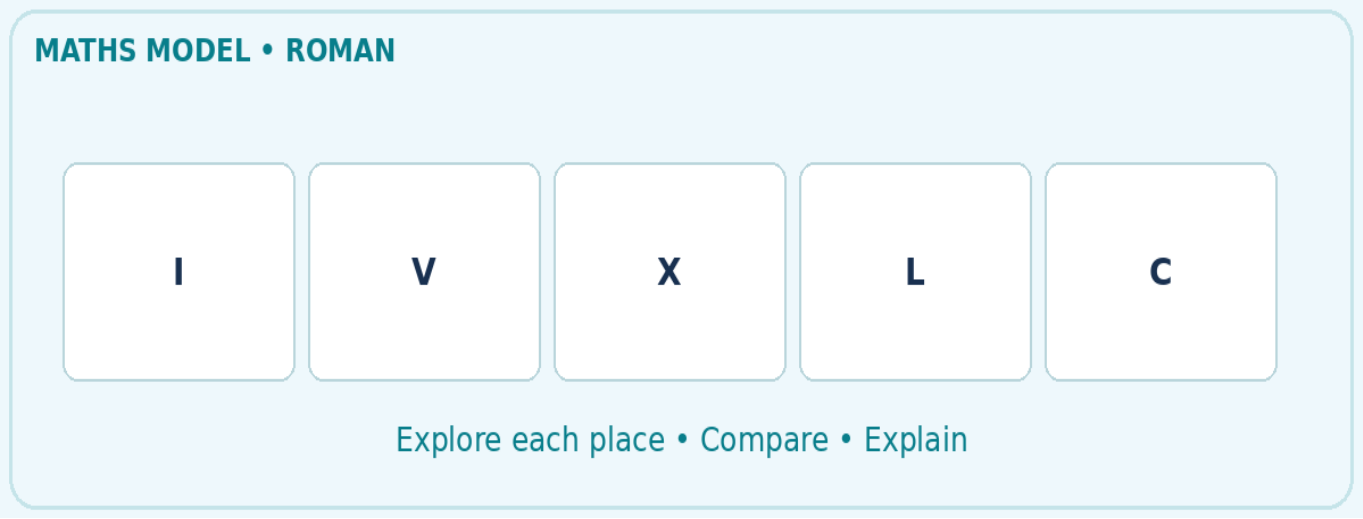
1. 40,000	2. 40,000	3. 48,000
4. 16,000	5. 35,000	6. 15,000
7. 56,000	8. 29,000	9. 20,000
10. 42,500	11. 20,000	12. 11,500
13. 20,000	14. 6,500	15. 42,500

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Recognising Roman numerals to C

LEARNING GOAL Recognising Roman numerals to C



LEARN THE IDEA

Roman symbols include i=1, v=5, x=10, l=50 and c=100. use iv, ix, xl and xc for subtraction forms.

WORKED EXAMPLE 1

Write 11 as a Roman numeral.

Answer: XI

Roman numerals use I, V, X, L and C: 11 = XI.

WORKED EXAMPLE 2

Convert LXXXIII into an ordinary number.

Answer: 83

LXXXIII represents 83.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Recognising Roman numerals to C

PLAY · LEARN THE STEPS

1. **Write 29 as a Roman numeral.**
   A) XXIX B) XLVIII C) LXVI D) XXXVII
2. **Convert LXIII into an ordinary number.**
   A) 68 B) 73 C) 63 D) 58
3. **Write 27 as a Roman numeral.**
   A) LXIV B) XXXV C) XLVI D) XXVII
4. **Convert LII into an ordinary number.**
   A) 62 B) 47 C) 57 D) 52
5. **Write 46 as a Roman numeral.**
   A) LXXXIII B) LIV C) XLVI D) LXV

CHALLENGE · DIFFERENT EXAMPLES

1. **A clock label for a school fundraiser is XLVII. What number is it?**
   A) 42 B) 47 C) 52 D) 57
2. **The award board at a farm shows position 24. Which Roman numeral is correct?**
   A) LXI B) XXXII C) XXIV D) XLIII
3. **A clock label during a sports day is XXXV. What number is it?**
   A) 35 B) 30 C) 40 D) 45
4. **The award board in a craft workshop shows position 34. Which Roman numeral is correct?**
   A) LXXI B) XLII C) XXXIV D) LIII
5. **A clock label at a school fair is XXV. What number is it?**
   A) 30 B) 25 C) 35 D) 20

Recognising Roman numerals to C

CHALLENGE · APPLY WHAT YOU KNOW

6. **The award board in the class garden shows position 91. Which Roman numeral is correct?**
   A) XCIX B) XCI C) XXVIII D) X

7. **A clock label at a market stall is VIII. What number is it?**
   A) 8 B) 3 C) 18 D) 13

8. **The award board during a library project shows position 79. Which Roman numeral is correct?**
   A) LXXXVII B) XCVIII C) LXXIX D) XVI

9. **A clock label on a football field is LXXXIII. What number is it?**
   A) 88 B) 93 C) 78 D) 83

10. **The award board in a recycling club shows position 46. Which Roman numeral is correct?**
    A) LIV B) LXV C) LXXXIII D) XLVI

ANSWER CHECK · LOOK ONLY AFTER TRYING

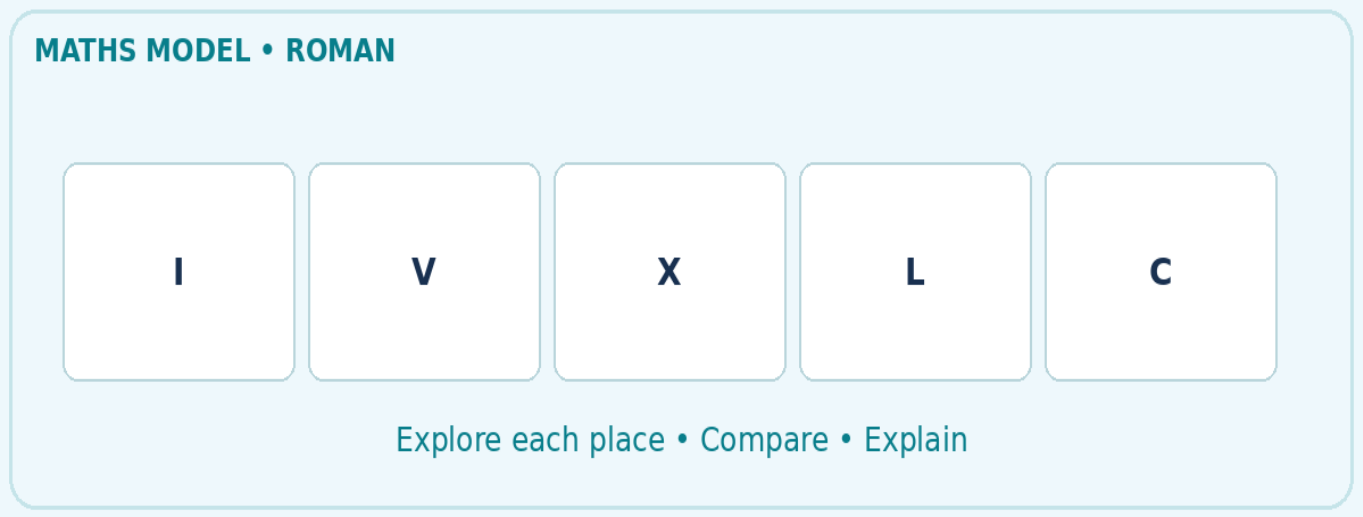
1. XXIX	2. 63	3. XXVII
4. 52	5. XLVI	6. 47
7. XXIV	8. 35	9. XXXIV
10. 25	11. XCI	12. 8
13. LXXIX	14. 83	15. XLVI

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Converting Roman and Arabic numerals

LEARNING GOAL Converting Roman and Arabic numerals



LEARN THE IDEA

Roman symbols include i=1, v=5, x=10, l=50 and c=100. use iv, ix, xl and xc for subtraction forms.

WORKED EXAMPLE 1

Write 64 as a Roman numeral.

Answer: LXIV

Roman numerals use I, V, X, L and C: 64 = LXIV.

WORKED EXAMPLE 2

Convert XXXIV into an ordinary number.

Answer: 34

XXXIV represents 34.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Converting Roman and Arabic numerals

PLAY · LEARN THE STEPS

1. **Write 40 as a Roman numeral.**
   A) XLVIII B) XL C) LXXVII D) LIX
2. **Convert XXVII into an ordinary number.**
   A) 27 B) 22 C) 37 D) 32
3. **Write 12 as a Roman numeral.**
   A) XII B) XX C) XLIX D) XXXI
4. **Convert XCV into an ordinary number.**
   A) 90 B) 95 C) 105 D) 100
5. **Write 23 as a Roman numeral.**
   A) LX B) XXXI C) XLII D) XXIII

CHALLENGE · DIFFERENT EXAMPLES

1. **A clock label at a farm is XLII. What number is it?**
   A) 37 B) 47 C) 42 D) 52
2. **The award board during a sports day shows position 2. Which Roman numeral is correct?**
   A) X B) II C) XXXIX D) XXI
3. **A clock label in a craft workshop is XXXVI. What number is it?**
   A) 36 B) 31 C) 41 D) 46
4. **The award board at a school fair shows position 43. Which Roman numeral is correct?**
   A) LXII B) LI C) LXXX D) XLIII
5. **A clock label in the class garden is XLV. What number is it?**
   A) 40 B) 50 C) 55 D) 45

Converting Roman and Arabic numerals

CHALLENGE · APPLY WHAT YOU KNOW

6. **The award board at a market stall shows position 21. Which Roman numeral is correct?**
   A) XXIX B) XL C) XXI D) LVIII

7. **A clock label during a library project is LII. What number is it?**
   A) 47 B) 57 C) 62 D) 52

8. **The award board on a football field shows position 76. Which Roman numeral is correct?**
   A) XCV B) LXXVI C) LXXXIV D) XIII

9. **A clock label in a recycling club is L. What number is it?**
   A) 55 B) 45 C) 60 D) 50

10. **The award board at a science exhibition shows position 31. Which Roman numeral is correct?**
    A) XXXI B) XXXIX C) LXVIII D) L

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. XL	2. 27	3. XII
4. 95	5. XXIII	6. 42
7. II	8. 36	9. XLIII
10. 45	11. XXI	12. 52
13. LXXVI	14. 50	15. XXXI

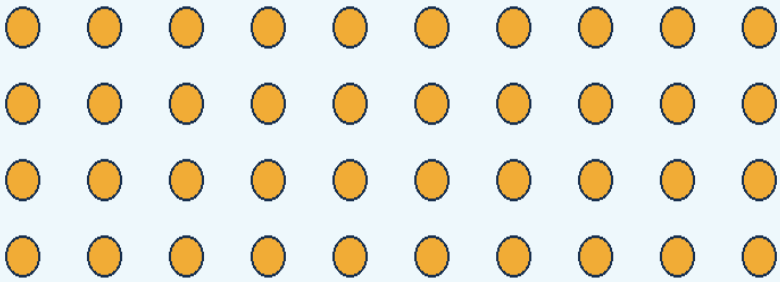
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Highest common factors by prime factors

LEARNING GOAL Highest common factors by prime factors

MATHS MODEL • HCF



4 groups of 10 = 40

LEARN THE IDEA

The highest common factor is the largest number that divides both numbers. factor trees and prime factors help.

WORKED EXAMPLE 1

Find HCF(30, 4).

Answer: 2

Highest common factor = 2.

WORKED EXAMPLE 2

Find HCF(12, 18).

Answer: 6

Highest common factor = 6.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Highest common factors by prime factors

PLAY · LEARN THE STEPS

1. **Find HCF(10, 15).**
   A) 4 B) 25 C) 5 D) 6
2. **Find HCF(6, 24).**
   A) 6 B) 5 C) 7 D) 30
3. **Find HCF(40, 6).**
   A) 2 B) 3 C) 46 D) 1
4. **Find HCF(20, 24).**
   A) 4 B) 44 C) 3 D) 5
5. **Find HCF(20, 14).**
   A) 3 B) 2 C) 34 D) 1

CHALLENGE · DIFFERENT EXAMPLES

1. **Two groups during a sports day hold 32 and 15 items. What is the greatest size of identical packets that divides both totals?**
   A) 1 B) 3 C) 47 D) 2
2. **Two groups in a craft workshop hold 10 and 16 items. What is the greatest size of identical packets that divides both totals?**
   A) 26 B) 3 C) 1 D) 2
3. **Two groups at a school fair hold 6 and 12 items. What is the greatest size of identical packets that divides both totals?**
   A) 18 B) 7 C) 5 D) 6
4. **Two groups in the class garden hold 12 and 4 items. What is the greatest size of identical packets that divides both totals?**
   A) 3 B) 16 C) 5 D) 4
5. **Two groups at a market stall hold 22 and 9 items. What is the greatest size of identical packets that divides both totals?**
   A) 3 B) 2 C) 31 D) 1

Highest common factors by prime factors

CHALLENGE · APPLY WHAT YOU KNOW

6 **Two groups during a library project hold 28 and 21 items. What is the greatest size of identical packets that divides both totals?**

A) 7 B) 8 C) 49 D) 6

7 **Two groups on a football field hold 60 and 9 items. What is the greatest size of identical packets that divides both totals?**

A) 2 B) 4 C) 3 D) 69

8 **Two groups in a recycling club hold 12 and 14 items. What is the greatest size of identical packets that divides both totals?**

A) 26 B) 1 C) 2 D) 3

9 **Two groups at a science exhibition hold 10 and 45 items. What is the greatest size of identical packets that divides both totals?**

A) 55 B) 4 C) 6 D) 5

10 **Two groups during a reading competition hold 18 and 14 items. What is the greatest size of identical packets that divides both totals?**

A) 32 B) 1 C) 2 D) 3

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 5	2. 6	3. 2
4. 4	5. 2	6. 1
7. 2	8. 6	9. 4
10. 1	11. 7	12. 3
13. 2	14. 5	15. 2

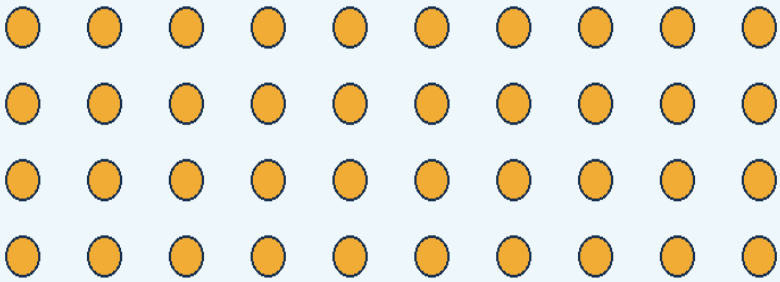
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Least common multiples by prime factors

LEARNING GOAL Least common multiples by prime factors

MATHS MODEL • LCM



4 groups of 10 = 40

LEARN THE IDEA

Multiples appear when you count by a number. the least common multiple is the first shared positive multiple.

WORKED EXAMPLE 1

Find LCM(12, 2).

Answer: 12

Least common multiple = 12.

WORKED EXAMPLE 2

Find LCM(7, 2).

Answer: 14

Least common multiple = 14.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Least common multiples by prime factors

PLAY · LEARN THE STEPS

1. **Find LCM(12, 6).**
   A) 12 B) 13 C) 11 D) 18
2. **Find LCM(2, 7).**
   A) 13 B) 9 C) 14 D) 15
3. **Find LCM(10, 5).**
   A) 10 B) 11 C) 15 D) 9
4. **Find LCM(6, 5).**
   A) 11 B) 30 C) 29 D) 31
5. **Find LCM(8, 3).**
   A) 24 B) 11 C) 23 D) 25

CHALLENGE · DIFFERENT EXAMPLES

1. **Two bells in a craft workshop ring every 8 and 5 minutes. After how many minutes will they ring together?**
   A) 41 B) 13 C) 39 D) 40
2. **Two bells at a school fair ring every 3 and 4 minutes. After how many minutes will they ring together?**
   A) 11 B) 13 C) 7 D) 12
3. **Two bells in the class garden ring every 10 and 6 minutes. After how many minutes will they ring together?**
   A) 16 B) 31 C) 29 D) 30
4. **Two bells at a market stall ring every 3 and 4 minutes. After how many minutes will they ring together?**
   A) 7 B) 12 C) 11 D) 13
5. **Two bells during a library project ring every 9 and 11 minutes. After how many minutes will they ring together?**
   A) 20 B) 98 C) 99 D) 100

Least common multiples by prime factors

CHALLENGE · APPLY WHAT YOU KNOW

6 **Two bells on a football field ring every 4 and 6 minutes. After how many minutes will they ring together?**

A) 10 B) 12 C) 13 D) 11

7 **Two bells in a recycling club ring every 12 and 5 minutes. After how many minutes will they ring together?**

A) 61 B) 17 C) 59 D) 60

8 **Two bells at a science exhibition ring every 6 and 7 minutes. After how many minutes will they ring together?**

A) 42 B) 43 C) 41 D) 13

9 **Two bells during a reading competition ring every 5 and 6 minutes. After how many minutes will they ring together?**

A) 30 B) 11 C) 31 D) 29

10 **Two bells for a school fundraiser ring every 6 and 5 minutes. After how many minutes will they ring together?**

A) 31 B) 30 C) 11 D) 29

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 12	2. 14	3. 10
4. 30	5. 24	6. 40
7. 12	8. 30	9. 12
10. 99	11. 12	12. 60
13. 42	14. 30	15. 30

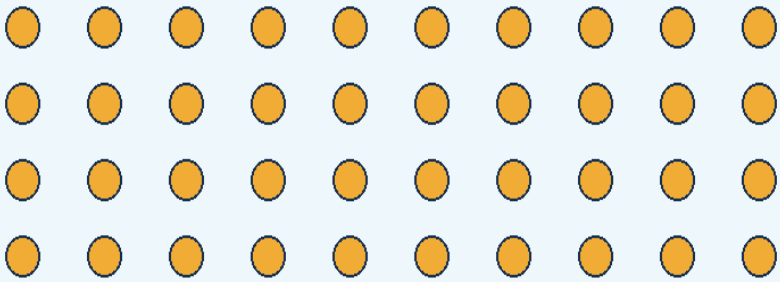
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Mental maths and number properties

LEARNING GOAL Mental maths and number properties

MATHS MODEL • MENTAL FACTS



4 groups of 10 = 40

LEARN THE IDEA

Build multiplication facts from known facts. for example, 6×7 = 5×7 + 7, so 35 + 7 = 42.

WORKED EXAMPLE 1

Use the known fact 6 × 8 = 48 to find 7 × 8.

Answer: 56

Add one group of 8: 48+8=56.

WORKED EXAMPLE 2

Use the known fact 6 × 2 = 12 to find 7 × 2.

Answer: 14

Add one group of 2: 12+2=14.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Mental maths and number properties

PLAY · LEARN THE STEPS

1. **Use the known fact 8 × 9 = 72 to find 9 × 9.**
   A) 81 B) 90 C) 18 D) 72
2. **Use the known fact 6 × 4 = 24 to find 7 × 4.**
   A) 24 B) 28 C) 32 D) 11
3. **Use the known fact 2 × 3 = 6 to find 3 × 3.**
   A) 9 B) 10 C) 12 D) 6
4. **Use the known fact 7 × 8 = 56 to find 8 × 8.**
   A) 16 B) 56 C) 72 D) 64
5. **Use the known fact 3 × 3 = 9 to find 4 × 3.**
   A) 12 B) 9 C) 7 D) 15

CHALLENGE · DIFFERENT EXAMPLES

1. **A learner at a school fair knows 4 × 5 = 20. How can they calculate 5 × 5 using one more group?**
   A) 10 B) 30 C) 20 D) 25
2. **A learner in the class garden knows 1 × 2 = 2. How can they calculate 2 × 2 using one more group?**
   A) 4 B) 5 C) 6 D) 2
3. **A learner at a market stall knows 2 × 5 = 10. How can they calculate 3 × 5 using one more group?**
   A) 15 B) 20 C) 8 D) 10
4. **A learner during a library project knows 8 × 2 = 16. How can they calculate 9 × 2 using one more group?**
   A) 11 B) 18 C) 16 D) 20
5. **A learner on a football field knows 2 × 6 = 12. How can they calculate 3 × 6 using one more group?**
   A) 9 B) 24 C) 12 D) 18

Mental maths and number properties

CHALLENGE · APPLY WHAT YOU KNOW

6. **A learner in a recycling club knows 8 × 8 = 64. How can they calculate 9 × 8 using one more group?**
   A) 64 B) 17 C) 72 D) 80

7. **A learner at a science exhibition knows 6 × 3 = 18. How can they calculate 7 × 3 using one more group?**
   A) 18 B) 10 C) 21 D) 24

8. **A learner during a reading competition knows 7 × 4 = 28. How can they calculate 8 × 4 using one more group?**
   A) 28 B) 12 C) 32 D) 36

9. **A learner for a school fundraiser knows 2 × 6 = 12. How can they calculate 3 × 6 using one more group?**
   A) 18 B) 9 C) 24 D) 12

10. **A learner at a farm knows 7 × 5 = 35. How can they calculate 8 × 5 using one more group?**
    A) 40 B) 35 C) 45 D) 13

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 81	2. 28	3. 9
4. 64	5. 12	6. 25
7. 4	8. 15	9. 18
10. 18	11. 72	12. 21
13. 32	14. 18	15. 40

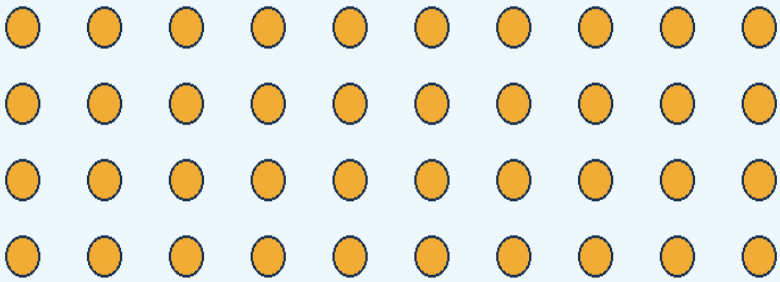
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Efficient mental multiplication

LEARNING GOAL Efficient mental multiplication

MATHS MODEL • MENTAL MULT



4 groups of 10 = 40

LEARN THE IDEA

Use place value and doubling: multiplying by 20 is multiplying by 2 and then by 10.

WORKED EXAMPLE 1

Work out 24 × 11 using an efficient method.

Answer: 264

24 × 11 = 264.

WORKED EXAMPLE 2

Work out 62 × 10 using an efficient method.

Answer: 620

62 × 10 = 620.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Efficient mental multiplication

PLAY · LEARN THE STEPS

1. **Work out 44 × 5 using an efficient method.**
   A) 230 B) 215 C) 220 D) 225
2. **Work out 22 × 100 using an efficient method.**
   A) 2210 B) 2300 C) 2100 D) 2200
3. **Work out 59 × 100 using an efficient method.**
   A) 5910 B) 6000 C) 5900 D) 5800
4. **Work out 63 × 11 using an efficient method.**
   A) 693 B) 703 C) 704 D) 682
5. **Work out 81 × 100 using an efficient method.**
   A) 8110 B) 8200 C) 8100 D) 8000

CHALLENGE · DIFFERENT EXAMPLES

1. **A project in the class garden has 37 units in each of 10 sets. What is the total?**
   A) 360 B) 370 C) 371 D) 380
2. **A project at a market stall has 97 units in each of 100 sets. What is the total?**
   A) 9800 B) 9710 C) 9700 D) 9600
3. **A project during a library project has 51 units in each of 10 sets. What is the total?**
   A) 511 B) 500 C) 520 D) 510
4. **A project on a football field has 77 units in each of 5 sets. What is the total?**
   A) 395 B) 390 C) 385 D) 380
5. **A project in a recycling club has 24 units in each of 10 sets. What is the total?**
   A) 230 B) 250 C) 240 D) 241

Efficient mental multiplication

CHALLENGE · APPLY WHAT YOU KNOW

6. **A project at a science exhibition has 51 units in each of 5 sets. What is the total?**
A) 250 B) 255 C) 265 D) 260

7. **A project during a reading competition has 75 units in each of 20 sets. What is the total?**
A) 1510 B) 1520 C) 1500 D) 1480

8. **A project for a school fundraiser has 97 units in each of 9 sets. What is the total?**
A) 873 B) 883 C) 882 D) 864

9. **A project at a farm has 49 units in each of 10 sets. What is the total?**
A) 490 B) 500 C) 480 D) 491

10. **A project during a sports day has 24 units in each of 5 sets. What is the total?**
A) 125 B) 115 C) 130 D) 120

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 220	2. 2200	3. 5900
4. 693	5. 8100	6. 370
7. 9700	8. 510	9. 385
10. 240	11. 255	12. 1500
13. 873	14. 490	15. 120

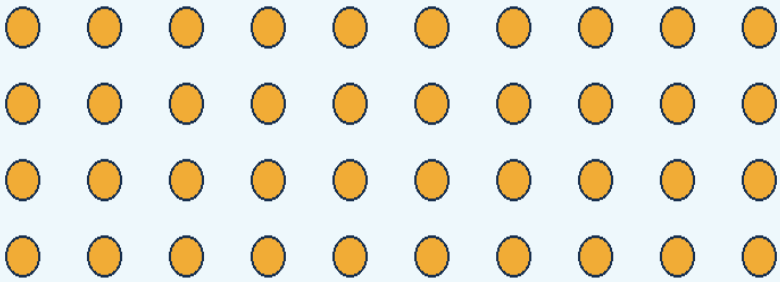
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Multiplication with multi-digit factors

LEARNING GOAL Multiplication with multi-digit factors

MATHS MODEL • MULTIPLY BIG



4 groups of 10 = 40

LEARN THE IDEA

Use partial products and place-value columns to multiply by a two-digit or three-digit factor. estimate to check.

WORKED EXAMPLE 1

Calculate 3,258 × 67.

Answer: 218,286

Use partial products or long multiplication to get 218,286.

WORKED EXAMPLE 2

Calculate 2,969 × 267.

Answer: 792,723

Use partial products or long multiplication to get 792,723.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Multiplication with multi-digit factors

PLAY · LEARN THE STEPS

1. **Calculate 2,552 × 47.**
   A) 2,599 B) 120,414 C) 122,496 D) 119,944
2. **Calculate 1,269 × 189.**
   A) 241,110 B) 239,841 C) 241,731 D) 1,458
3. **Calculate 2,126 × 71.**
   A) 150,946 B) 153,072 C) 151,656 D) 2,197
4. **Calculate 1,324 × 245.**
   A) 326,830 B) 324,380 C) 1,569 D) 325,704
5. **Calculate 7,262 × 32.**
   A) 239,646 B) 232,704 C) 7,294 D) 232,384

CHALLENGE · DIFFERENT EXAMPLES

1. **A supplier in a recycling club delivers 290 cartons with 6290 pencils in each. How many pencils altogether?**
   A) 1,830,390 B) 1,824,100 C) 1,827,000 D) 6,580
2. **A supplier at a science exhibition delivers 79 cartons with 1608 pencils in each. How many pencils altogether?**
   A) 128,640 B) 127,822 C) 1,687 D) 127,032
3. **A supplier during a reading competition delivers 255 cartons with 6704 pencils in each. How many pencils altogether?**
   A) 1,716,224 B) 1,709,520 C) 1,712,070 D) 6,959
4. **A supplier for a school fundraiser delivers 26 cartons with 3338 pencils in each. How many pencils altogether?**
   A) 87,048 B) 3,364 C) 86,788 D) 90,126
5. **A supplier at a farm delivers 229 cartons with 6811 pencils in each. How many pencils altogether?**
   A) 1,559,719 B) 1,566,530 C) 1,562,009 D) 7,040

Multiplication with multi-digit factors

CHALLENGE · APPLY WHAT YOU KNOW

6 **A supplier during a sports day delivers 51 cartons with 2451 pencils in each. How many pencils altogether?**

A) 125,001 B) 127,452 C) 125,511 D) 2,502

7 **A supplier in a craft workshop delivers 275 cartons with 2738 pencils in each. How many pencils altogether?**

A) 752,950 B) 755,688 C) 3,013 D) 755,700

8 **A supplier at a school fair delivers 69 cartons with 3583 pencils in each. How many pencils altogether?**

A) 250,810 B) 247,917 C) 247,227 D) 3,652

9 **A supplier in the class garden delivers 184 cartons with 3901 pencils in each. How many pencils altogether?**

A) 719,624 B) 721,685 C) 717,784 D) 4,085

10 **A supplier at a market stall delivers 29 cartons with 5423 pencils in each. How many pencils altogether?**

A) 162,690 B) 157,557 C) 157,267 D) 5,452

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 119,944	2. 239,841	3. 150,946
4. 324,380	5. 232,384	6. 1,824,100
7. 127,032	8. 1,709,520	9. 86,788
10. 1,559,719	11. 125,001	12. 752,950
13. 247,227	14. 717,784	15. 157,267

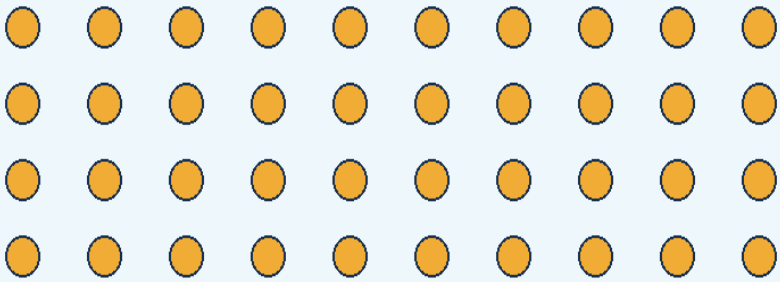
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Division fact families

LEARNING GOAL Division fact families

MATHS MODEL • DIVISION FACTS



4 groups of 10 = 40

LEARN THE IDEA

Division undoes multiplication: if 7×8=56, then 56÷7=8 and 56÷8=7.

WORKED EXAMPLE 1

Calculate 6 ÷ 2.

Answer: 3

2 × 3 = 6.

WORKED EXAMPLE 2

Calculate 30 ÷ 6.

Answer: 5

6 × 5 = 30.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Division fact families

PLAY · LEARN THE STEPS

1. **Calculate 45 ÷ 9.**
   A) 36 B) 5 C) 6 D) 9
2. **Calculate 36 ÷ 6.**
   A) 6 B) 30 C) 8 D) 7
3. **Calculate 15 ÷ 5.**
   A) 5 B) 3 C) 10 D) 4
4. **Calculate 18 ÷ 9.**
   A) 4 B) 3 C) 9 D) 2
5. **Calculate 20 ÷ 4.**
   A) 16 B) 4 C) 5 D) 6

CHALLENGE · DIFFERENT EXAMPLES

1. **A club during a library project shares 25 pencils equally among 5 teams. How many per team?**
   A) 5 B) 20 C) 7 D) 6
2. **A club on a football field shares 18 pencils equally among 6 teams. How many per team?**
   A) 12 B) 4 C) 3 D) 6
3. **A club in a recycling club shares 28 pencils equally among 4 teams. How many per team?**
   A) 4 B) 8 C) 24 D) 7
4. **A club at a science exhibition shares 32 pencils equally among 4 teams. How many per team?**
   A) 4 B) 8 C) 28 D) 9
5. **A club during a reading competition shares 21 pencils equally among 7 teams. How many per team?**
   A) 4 B) 14 C) 3 D) 7

Division fact families

CHALLENGE · APPLY WHAT YOU KNOW

6. **A club for a school fundraiser shares 45 pencils equally among 9 teams. How many per team?**
   A) 6 B) 9 C) 36 D) 5

7. **A club at a farm shares 18 pencils equally among 9 teams. How many per team?**
   A) 2 B) 4 C) 9 D) 3

8. **A club during a sports day shares 6 pencils equally among 2 teams. How many per team?**
   A) 2 B) 5 C) 3 D) 4

9. **A club in a craft workshop shares 14 pencils equally among 7 teams. How many per team?**
   A) 4 B) 7 C) 3 D) 2

10. **A club at a school fair shares 72 pencils equally among 9 teams. How many per team?**
    A) 9 B) 10 C) 8 D) 63

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 5	2. 6	3. 3
4. 2	5. 5	6. 5
7. 3	8. 7	9. 8
10. 3	11. 5	12. 2
13. 3	14. 2	15. 8

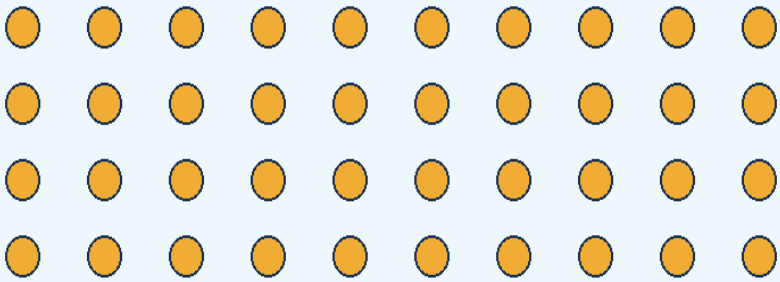
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Three-digit division

LEARNING GOAL Three-digit division

MATHS MODEL • LONG DIVISION



4 groups of 10 = 40

LEARN THE IDEA

Divide hundreds, then tens, then ones. check every answer by multiplication.

WORKED EXAMPLE 1

Calculate 368 ÷ 8.

Answer: 46

8 × 46 = 368.

WORKED EXAMPLE 2

Calculate 395 ÷ 5.

Answer: 79

5 × 79 = 395.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Three-digit division

PLAY · LEARN THE STEPS

1. **Calculate 455 ÷ 5.**
   A) 91 B) 450 C) 5 D) 92
2. **Calculate 117 ÷ 3.**
   A) 114 B) 39 C) 40 D) 3
3. **Calculate 560 ÷ 7.**
   A) 553 B) 7 C) 80 D) 81
4. **Calculate 96 ÷ 3.**
   A) 3 B) 93 C) 32 D) 33
5. **Calculate 145 ÷ 5.**
   A) 29 B) 140 C) 30 D) 5

CHALLENGE · DIFFERENT EXAMPLES

1. **A club on a football field shares 450 pencils equally among 6 teams. How many per team?**
   A) 6 B) 75 C) 444 D) 76
2. **A club in a recycling club shares 472 pencils equally among 8 teams. How many per team?**
   A) 464 B) 59 C) 60 D) 8
3. **A club at a science exhibition shares 360 pencils equally among 5 teams. How many per team?**
   A) 355 B) 73 C) 5 D) 72
4. **A club during a reading competition shares 464 pencils equally among 8 teams. How many per team?**
   A) 59 B) 8 C) 58 D) 456
5. **A club for a school fundraiser shares 261 pencils equally among 9 teams. How many per team?**
   A) 252 B) 30 C) 9 D) 29

Three-digit division

CHALLENGE · APPLY WHAT YOU KNOW

6. **A club at a farm shares 385 pencils equally among 7 teams. How many per team?**
   A) 7 B) 55 C) 378 D) 56

7. **A club during a sports day shares 150 pencils equally among 3 teams. How many per team?**
   A) 3 B) 50 C) 147 D) 51

8. **A club in a craft workshop shares 144 pencils equally among 4 teams. How many per team?**
   A) 36 B) 37 C) 140 D) 4

9. **A club at a school fair shares 126 pencils equally among 7 teams. How many per team?**
   A) 19 B) 18 C) 119 D) 7

10. **A club in the class garden shares 340 pencils equally among 4 teams. How many per team?**
    A) 86 B) 336 C) 85 D) 4

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 91	2. 39	3. 80
4. 32	5. 29	6. 75
7. 59	8. 72	9. 58
10. 29	11. 55	12. 50
13. 36	14. 18	15. 85

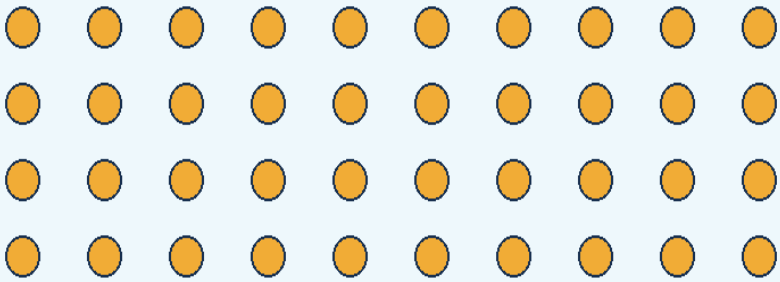
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Multi-step problems with four operations

LEARNING GOAL Multi-step problems with four operations

MATHS MODEL • WORD PROBLEMS



4 groups of 10 = 40

LEARN THE IDEA

Read the question carefully, list the needed quantities, choose operations in order, then check units.

WORKED EXAMPLE 1

First add 15 and 41, then multiply the result by 5.

Answer: 280

(15+41) × 5 = 280.

WORKED EXAMPLE 2

First add 16 and 14, then multiply the result by 2.

Answer: 60

(16+14) × 2 = 60.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Multi-step problems with four operations

PLAY · LEARN THE STEPS

1. **First add 20 and 18, then multiply the result by 5.**
   A) 190 B) 43 C) 195 D) 38
2. **First add 22 and 39, then multiply the result by 4.**
   A) 244 B) 61 C) 248 D) 65
3. **First add 20 and 39, then multiply the result by 5.**
   A) 295 B) 64 C) 300 D) 59
4. **First add 24 and 32, then multiply the result by 2.**
   A) 114 B) 112 C) 58 D) 56
5. **First add 38 and 26, then multiply the result by 3.**
   A) 192 B) 195 C) 67 D) 64

CHALLENGE · DIFFERENT EXAMPLES

1. **A classroom in a recycling club has 32 blue counters and 42 red counters in each box. There are 2 boxes. How many counters altogether?**
   A) 74 B) 150 C) 148 D) 76
2. **A classroom at a science exhibition has 38 blue counters and 21 red counters in each box. There are 5 boxes. How many counters altogether?**
   A) 64 B) 295 C) 300 D) 59
3. **A classroom during a reading competition has 32 blue counters and 13 red counters in each box. There are 5 boxes. How many counters altogether?**
   A) 45 B) 225 C) 230 D) 50
4. **A classroom for a school fundraiser has 28 blue counters and 12 red counters in each box. There are 4 boxes. How many counters altogether?**
   A) 160 B) 164 C) 44 D) 40
5. **A classroom at a farm has 16 blue counters and 40 red counters in each box. There are 3 boxes. How many counters altogether?**
   A) 171 B) 59 C) 168 D) 56

Multi-step problems with four operations

CHALLENGE · APPLY WHAT YOU KNOW

6 **A classroom during a sports day has 37 blue counters and 47 red counters in each box. There are 5 boxes. How many counters altogether?**

A) 425 B) 84 C) 89 D) 420

7 **A classroom in a craft workshop has 12 blue counters and 18 red counters in each box. There are 5 boxes. How many counters altogether?**

A) 155 B) 35 C) 30 D) 150

8 **A classroom at a school fair has 13 blue counters and 26 red counters in each box. There are 3 boxes. How many counters altogether?**

A) 117 B) 42 C) 39 D) 120

9 **A classroom in the class garden has 29 blue counters and 45 red counters in each box. There are 4 boxes. How many counters altogether?**

A) 300 B) 78 C) 74 D) 296

10 **A classroom at a market stall has 16 blue counters and 20 red counters in each box. There are 4 boxes. How many counters altogether?**

A) 36 B) 40 C) 148 D) 144

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 190	2. 244	3. 295
4. 112	5. 192	6. 148
7. 295	8. 225	9. 160
10. 168	11. 420	12. 150
13. 117	14. 296	15. 144

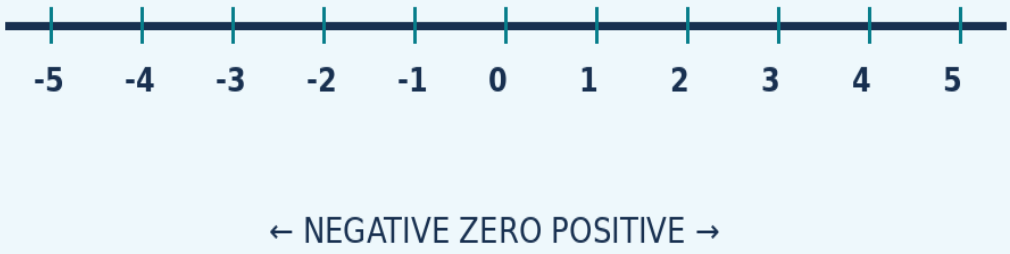
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Locating and comparing integers

LEARNING GOAL Locating and comparing integers

MATHS MODEL • INTEGER COMPARE



LEARN THE IDEA

Negative numbers lie to the left of zero; the number farther right is greater, even if it has fewer digits.

WORKED EXAMPLE 1

Compare the integers: -9 __ 19.

Answer: <

On a number line, -9 is to the left of 19.

WORKED EXAMPLE 2

Compare the integers: -38 __ 33.

Answer: <

On a number line, -38 is to the left of 33.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Locating and comparing integers

PLAY · LEARN THE STEPS

1. **Compare the integers: -22 __ 24.**
   A) none of these B) > C) = D) <
2. **Compare the integers: 13 __ 24.**
   A) = B) > C) < D) none of these
3. **Compare the integers: 18 __ -5.**
   A) < B) = C) none of these D) >
4. **Compare the integers: 2 __ -27.**
   A) > B) none of these C) = D) <
5. **Compare the integers: 0 __ 3.**
   A) < B) = C) none of these D) >

CHALLENGE · DIFFERENT EXAMPLES

1. **At a weather station at a farm, two readings are -1°C and 26°C. Which sign makes -1 __ 26 true?**
   A) < B) = C) > D) none of these
2. **At a weather station during a sports day, two readings are 2°C and 3°C. Which sign makes 2 __ 3 true?**
   A) = B) > C) none of these D) <
3. **At a weather station in a craft workshop, two readings are -15°C and 6°C. Which sign makes -15 __ 6 true?**
   A) none of these B) < C) = D) >
4. **At a weather station at a school fair, two readings are -9°C and -18°C. Which sign makes -9 __ -18 true?**
   A) = B) < C) > D) none of these
5. **At a weather station in the class garden, two readings are -39°C and -8°C. Which sign makes -39 __ -8 true?**
   A) none of these B) = C) < D) >

Locating and comparing integers

CHALLENGE · APPLY WHAT YOU KNOW

6 **At a weather station at a market stall, two readings are -6°C and 12°C. Which sign makes -6 __ 12 true?**

A) none of these B) > C) = D) <

7 **At a weather station during a library project, two readings are -39°C and 19°C. Which sign makes -39 __ 19 true?**

A) < B) none of these C) > D) =

8 **At a weather station on a football field, two readings are -35°C and 17°C. Which sign makes -35 __ 17 true?**

A) < B) > C) = D) none of these

9 **At a weather station in a recycling club, two readings are 13°C and 18°C. Which sign makes 13 __ 18 true?**

A) none of these B) = C) < D) >

10 **At a weather station at a science exhibition, two readings are -30°C and 18°C. Which sign makes -30 __ 18 true?**

A) > B) < C) = D) none of these

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. <	2. <	3. >
4. >	5. <	6. <
7. <	8. <	9. >
10. <	11. <	12. <
13. <	14. <	15. <

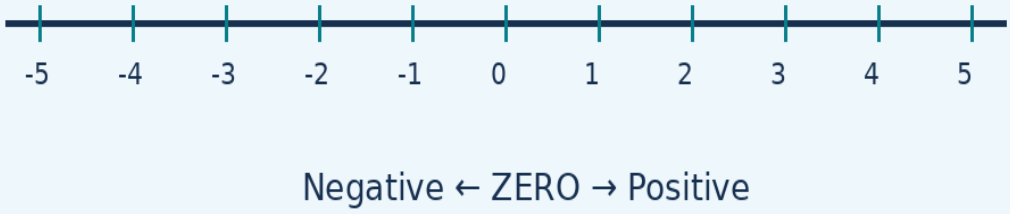
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Adding and subtracting integers

LEARNING GOAL Adding and subtracting integers

MATHS MODEL • INTEGER



Negative ← ZERO → Positive

LEARN THE IDEA

Integers include positive and negative numbers. on the number line right adds and left subtracts.

WORKED EXAMPLE 1

Calculate -7 + (-8).

Answer: -15

On the number line, start at -7 and move -8.

WORKED EXAMPLE 2

Calculate 14 + (0).

Answer: 14

On the number line, start at 14 and move +0.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Adding and subtracting integers

PLAY · LEARN THE STEPS

1. **Calculate 9 + (-4).**
   A) 7 B) 5 C) 13 D) 3
2. **Calculate -8 + (-6).**
   A) -2 B) -16 C) -14 D) -12
3. **Calculate 7 + (-6).**
   A) -1 B) 13 C) 3 D) 1
4. **Calculate -2 + (-2).**
   A) -4 B) -6 C) -2 D) 0
5. **Calculate -8 + (0).**
   A) -10 B) -6 C) -8 D) -7

CHALLENGE · DIFFERENT EXAMPLES

1. **The thermometer during a reading competition reads 0°C and changes by -1°C. What does it read afterward?**
   A) 0 B) 1 C) -3 D) -1
2. **The thermometer for a school fundraiser reads 0°C and changes by -4°C. What does it read afterward?**
   A) -4 B) -2 C) -6 D) 4
3. **The thermometer at a farm reads -9°C and changes by -1°C. What does it read afterward?**
   A) -9 B) -12 C) -8 D) -10
4. **The thermometer during a sports day reads -6°C and changes by -2°C. What does it read afterward?**
   A) -10 B) -4 C) -8 D) -6
5. **The thermometer in a craft workshop reads 9°C and changes by -10°C. What does it read afterward?**
   A) 19 B) 1 C) -1 D) -3

Adding and subtracting integers

CHALLENGE · APPLY WHAT YOU KNOW

6. **The thermometer at a school fair reads 12°C and changes by +8°C. What does it read afterward?**

   A) 4 B) 20 C) 22 D) 18

7. **The thermometer in the class garden reads -13°C and changes by +4°C. What does it read afterward?**

   A) -7 B) -9 C) -11 D) -17

8. **The thermometer at a market stall reads 7°C and changes by -7°C. What does it read afterward?**

   A) 0 B) 2 C) 14 D) -2

9. **The thermometer during a library project reads -4°C and changes by +3°C. What does it read afterward?**

   A) -7 B) -1 C) -3 D) 1

10. **The thermometer on a football field reads -14°C and changes by -3°C. What does it read afterward?**

    A) -17 B) -11 C) -15 D) -19

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 5	2. -14	3. 1
4. -4	5. -8	6. -1
7. -4	8. -10	9. -8
10. -1	11. 20	12. -9
13. 0	14. -1	15. -17

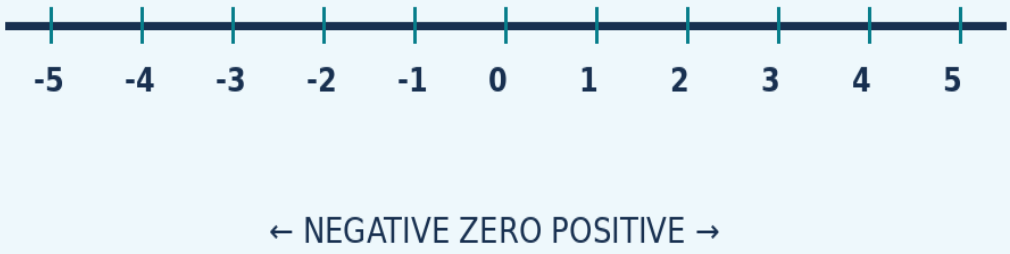
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Multiplication with positive and negative integers

LEARNING GOAL Multiplication with positive and negative integers

MATHS MODEL • INTEGER MULT



LEARN THE IDEA

A positive times a negative gives a negative; two negatives multiply to a positive.

WORKED EXAMPLE 1

Calculate 11 × (6).

Answer: 66

Equal signs produce positive; different signs produce negative. Result: 66.

WORKED EXAMPLE 2

Calculate -10 × (-8).

Answer: 80

Equal signs produce positive; different signs produce negative. Result: 80.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Multiplication with positive and negative integers

PLAY · LEARN THE STEPS

1. **Calculate 4 × (5).**
   A) 20 B) -20 C) 21 D) 24
2. **Calculate -10 × (-3).**
   A) 30 B) 31 C) 40 D) -30
3. **Calculate 6 × (-4).**
   A) -23 B) 24 C) -24 D) -18
4. **Calculate -9 × (4).**
   A) -36 B) -27 C) 36 D) -35
5. **Calculate 5 × (-2).**
   A) -9 B) -5 C) 10 D) -10

CHALLENGE · DIFFERENT EXAMPLES

1. **A number-line challenge in a craft workshop asks for the product of two signed numbers: -3 × (-7). What is the result?**
   A) 22 B) 21 C) 24 D) -21
2. **A number-line challenge at a school fair asks for the product of two signed numbers: 8 × (-9). What is the result?**
   A) -71 B) -72 C) -64 D) 72
3. **A number-line challenge in the class garden asks for the product of two signed numbers: -11 × (2). What is the result?**
   A) -11 B) -22 C) -21 D) 22
4. **A number-line challenge at a market stall asks for the product of two signed numbers: 4 × (-9). What is the result?**
   A) 36 B) -36 C) -35 D) -32
5. **A number-line challenge during a library project asks for the product of two signed numbers: -6 × (-5). What is the result?**
   A) 36 B) 31 C) 30 D) -30

Multiplication with positive and negative integers

CHALLENGE · APPLY WHAT YOU KNOW

6. **A number-line challenge on a football field asks for the product of two signed numbers: 5 × (6). What is the result?**
   A) 35 B) -30 C) 31 D) 30

7. **A number-line challenge in a recycling club asks for the product of two signed numbers: -3 × (-8). What is the result?**
   A) 25 B) -24 C) 24 D) 27

8. **A number-line challenge at a science exhibition asks for the product of two signed numbers: 11 × (-4). What is the result?**
   A) -33 B) 44 C) -44 D) -43

9. **A number-line challenge during a reading competition asks for the product of two signed numbers: -9 × (7). What is the result?**
   A) -54 B) -62 C) 63 D) -63

10. **A number-line challenge for a school fundraiser asks for the product of two signed numbers: 9 × (-2). What is the result?**
    A) 18 B) -18 C) -17 D) -9

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 20	2. 30	3. -24
4. -36	5. -10	6. 21
7. -72	8. -22	9. -36
10. 30	11. 30	12. 24
13. -44	14. -63	15. -18

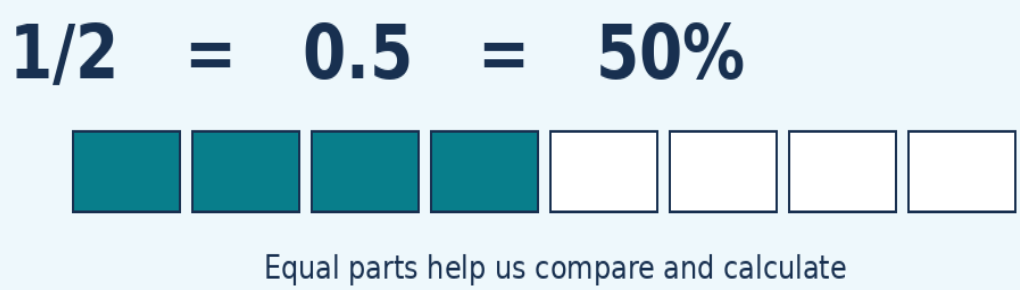
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Comparing fractions, decimals and percentages

LEARNING GOAL Comparing fractions, decimals and percentages

MATHS MODEL • MIXED COMPARE



Equal parts help us compare and calculate

LEARN THE IDEA

Translate fractions, decimals and percentages into a common representation before comparing.

WORKED EXAMPLE 1

Express 1/5 as a decimal.

Answer: 0.2

Calculate numerator ÷ denominator: 1 ÷ 5 = 0.2.

WORKED EXAMPLE 2

Convert 7/10 to a percentage.

Answer: 70%

7/10 = 0.7 = 70%.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Comparing fractions, decimals and percentages

PLAY · LEARN THE STEPS

1. **Express 2/5 as a decimal.**
   A) 0.5 B) 0.3 C) 0.4 D) 2.5
2. **Convert 3/5 to a percentage.**
   A) 60% B) 5% C) 70% D) 50%
3. **Express 1/10 as a decimal.**
   A) 0.2 B) 0.1 C) 10 D) 0
4. **Convert 1/2 to a percentage.**
   A) 2% B) 60% C) 50% D) 40%
5. **Express 1/2 as a decimal.**
   A) 0.5 B) 0.6 C) 0.4 D) 2

CHALLENGE · DIFFERENT EXAMPLES

1. **A learner at a school fair has completed 3/10 of a project. What percentage is equal to that fraction?**
   A) 20% B) 40% C) 10% D) 30%
2. **Which decimal is equivalent to 50% of a whole in the class garden?**
   A) 0.5 B) 0.4 C) 2 D) 0.6
3. **A learner at a market stall has completed 2/4 of a project. What percentage is equal to that fraction?**
   A) 4% B) 60% C) 50% D) 40%
4. **Which decimal is equivalent to 80% of a whole during a library project?**
   A) 0.9 B) 0.8 C) 0.7 D) 1.25
5. **A learner on a football field has completed 2/4 of a project. What percentage is equal to that fraction?**
   A) 60% B) 4% C) 40% D) 50%

Comparing fractions, decimals and percentages

CHALLENGE · APPLY WHAT YOU KNOW

6. **Which decimal is equivalent to 50% of a whole in a recycling club?**
A) 2 B) 0.5 C) 0.6 D) 0.4

7. **A learner at a science exhibition has completed 9/10 of a project. What percentage is equal to that fraction?**
A) 10% B) 100% C) 80% D) 90%

8. **Which decimal is equivalent to 80% of a whole during a reading competition?**
A) 0.7 B) 0.9 C) 0.8 D) 1.25

9. **A learner for a school fundraiser has completed 1/2 of a project. What percentage is equal to that fraction?**
A) 60% B) 40% C) 2% D) 50%

10. **Which decimal is equivalent to 50% of a whole at a farm?**
A) 2 B) 0.5 C) 0.4 D) 0.6

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 0.4	2. 60%	3. 0.1
4. 50%	5. 0.5	6. 30%
7. 0.5	8. 50%	9. 0.8
10. 50%	11. 0.5	12. 90%
13. 0.8	14. 50%	15. 0.5

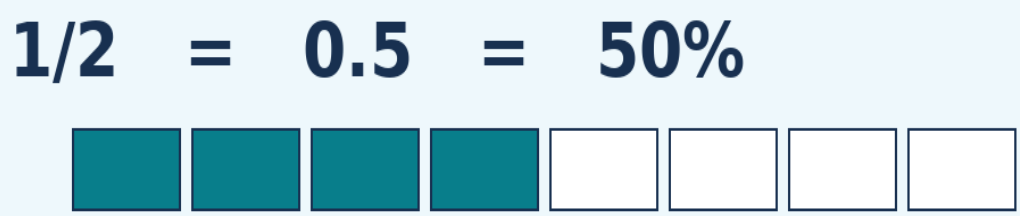
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Adding and subtracting unlike fractions

LEARNING GOAL Adding and subtracting unlike fractions

MATHS MODEL • FRACTION UNLIKE



Equal parts help us compare and calculate

LEARN THE IDEA

Unlike fractions have different denominator-sized parts. find a common denominator before combining them.

WORKED EXAMPLE 1

Work out 2/3 − 3/10.

Answer: 11/30

Use a common denominator, combine numerators and simplify to 11/30.

WORKED EXAMPLE 2

Work out 1/2 + 1/2.

Answer: 1/1

Use a common denominator, combine numerators and simplify to 1/1.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Adding and subtracting unlike fractions

PLAY · LEARN THE STEPS

1. **Work out 3/4 − 1/2.**
   A) 1/4 B) 1/2 C) 1/5 D) 3/4
2. **Work out 1/2 + 1/4.**
   A) 1/1 B) 3/5 C) 3/4 D) 1/2
3. **Work out 3/5 − 5/12.**
   A) 11/61 B) 3/5 C) 11/60 D) 1/5
4. **Work out 1/4 + 5/12.**
   A) 1/2 B) 1/4 C) 1/1 D) 2/3
5. **Work out 2/3 − 3/8.**
   A) 1/3 B) 7/24 C) 7/25 D) 2/3

CHALLENGE · DIFFERENT EXAMPLES

1. **Two measured lengths in the class garden are 1/2 m and 1/4 m. Find their total.**
   A) 1/1 B) 1/2 C) 3/4 D) 3/5
2. **Two measured lengths at a market stall are 1/2 m and 1/4 m. Find their positive difference.**
   A) 1/5 B) 1/4 C) 2/4 D) 1/2
3. **Two measured lengths during a library project are 4/5 m and 1/2 m. Find their total.**
   A) 13/10 B) 4/5 C) 7/5 D) 13/11
4. **Two measured lengths on a football field are 3/5 m and 2/5 m. Find their positive difference.**
   A) 2/5 B) 3/5 C) 1/5 D) 1/6
5. **Two measured lengths in a recycling club are 2/3 m and 5/6 m. Find their total.**
   A) 3/2 B) 1/1 C) 2/1 D) 2/3

Adding and subtracting unlike fractions

CHALLENGE · APPLY WHAT YOU KNOW

6 **Two measured lengths at a science exhibition are 1/3 m and 1/3 m. Find their positive difference.**

A) 1/1 B) 1/3 C) 0/2 D) 0/1

7 **Two measured lengths during a reading competition are 1/2 m and 1/3 m. Find their total.**

A) 1/2 B) 5/6 C) 5/7 D) 1/1

8 **Two measured lengths for a school fundraiser are 1/2 m and 1/4 m. Find their positive difference.**

A) 1/5 B) 1/4 C) 1/2 D) 2/4

9 **Two measured lengths at a farm are 1/3 m and 5/6 m. Find their total.**

A) 4/3 B) 7/6 C) 1/1 D) 1/3

10 **Two measured lengths during a sports day are 1/2 m and 1/2 m. Find their positive difference.**

A) 1/1 B) 1/2 C) 0/1 D) 0/2

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 1/4	2. 3/4	3. 11/60
4. 2/3	5. 7/24	6. 3/4
7. 1/4	8. 13/10	9. 1/5
10. 3/2	11. 0/1	12. 5/6
13. 1/4	14. 7/6	15. 0/1

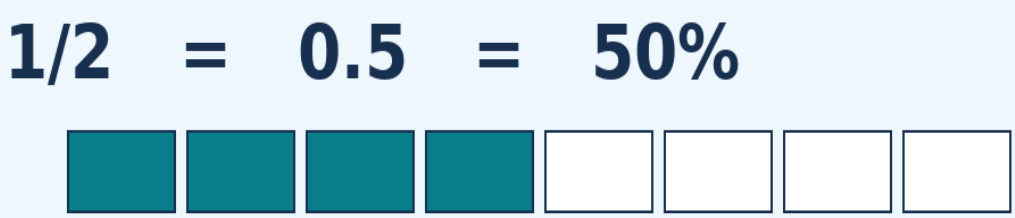
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Multiplying fractions by whole numbers

LEARNING GOAL Multiplying fractions by whole numbers

MATHS MODEL • FRACTION WHOLE



Equal parts help us compare and calculate

LEARN THE IDEA

Multiplying a fraction by a whole number means several equal fractional groups; use strips to model.

WORKED EXAMPLE 1

Calculate 1/5 of 35.

Answer: 7

One 5th of 35 is 7; multiply by 1 to get 7.

WORKED EXAMPLE 2

Calculate 1/2 of 16.

Answer: 8

One 2th of 16 is 8; multiply by 1 to get 8.

TRY THIS IN REAL LIFE

Cut or colour equal paper strips; explain each numerator and denominator before calculating.

Multiplying fractions by whole numbers

PLAY · LEARN THE STEPS

1. **Calculate 2/8 of 32.**
   A) 32 B) 10 C) 4 D) 8
2. **Calculate 2/3 of 12.**
   A) 10 B) 4 C) 12 D) 8
3. **Calculate 2/4 of 8.**
   A) 8 B) 4 C) 2 D) 6
4. **Calculate 1/2 of 12.**
   A) 7 B) 12 C) 8 D) 6
5. **Calculate 1/5 of 45.**
   A) 10 B) 11 C) 45 D) 9

CHALLENGE · DIFFERENT EXAMPLES

1. **4/6 of 24 reading cards are used in a craft workshop. How many cards is that?**
   A) 20 B) 4 C) 16 D) 24
2. **4/6 of 30 reading cards are used at a school fair. How many cards is that?**
   A) 30 B) 24 C) 5 D) 20
3. **1/6 of 12 reading cards are used in the class garden. How many cards is that?**
   A) 4 B) 2 C) 3 D) 12
4. **1/4 of 36 reading cards are used at a market stall. How many cards is that?**
   A) 11 B) 10 C) 9 D) 36
5. **5/8 of 40 reading cards are used during a library project. How many cards is that?**
   A) 30 B) 40 C) 25 D) 5

Multiplying fractions by whole numbers

CHALLENGE · APPLY WHAT YOU KNOW

6. **2/3 of 6 reading cards are used on a football field. How many cards is that?**
A) 4 B) 6 C) 2 D) 5

7. **1/2 of 6 reading cards are used in a recycling club. How many cards is that?**
A) 3 B) 5 C) 4 D) 6

8. **2/3 of 12 reading cards are used at a science exhibition. How many cards is that?**
A) 4 B) 12 C) 8 D) 10

9. **3/4 of 32 reading cards are used during a reading competition. How many cards is that?**
A) 24 B) 27 C) 32 D) 8

10. **1/2 of 6 reading cards are used for a school fundraiser. How many cards is that?**
A) 3 B) 4 C) 6 D) 5

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 8	2. 8	3. 4
4. 6	5. 9	6. 16
7. 20	8. 2	9. 9
10. 25	11. 4	12. 3
13. 8	14. 24	15. 3

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Representing ratios with objects

LEARNING GOAL Representing ratios with objects

MATHS MODEL • RATIO MODEL



$$3 : 5 = 6 : 10$$

Scale BOTH parts by the same number

LEARN THE IDEA

A ratio compares quantities in the same order; 2:3 means two units of the first for every three of the second.

WORKED EXAMPLE 1

Simplify the ratio 8:10.

Answer: 4:5

Divide both parts by the common factor. The ratio is 4:5.

WORKED EXAMPLE 2

Simplify the ratio 2:10.

Answer: 1:5

Divide both parts by the common factor. The ratio is 1:5.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Representing ratios with objects

PLAY · LEARN THE STEPS

1. **Simplify the ratio 8:10.**
   A) 4:5 B) 5:4 C) 9:5 D) 4:9
2. **Simplify the ratio 18:20.**
   A) 9:10 B) 10:9 C) 9:19 D) 19:10
3. **Simplify the ratio 10:16.**
   A) 5:8 B) 5:13 C) 13:8 D) 8:5
4. **Simplify the ratio 16:10.**
   A) 8:13 B) 13:5 C) 5:8 D) 8:5
5. **Simplify the ratio 6:8.**
   A) 4:3 B) 3:7 C) 7:4 D) 3:4

CHALLENGE · DIFFERENT EXAMPLES

1. **A bead string during a library project has 9 blue and 6 yellow beads. What is the simplified blue:yellow ratio?**
   A) 2:3 B) 5:2 C) 3:5 D) 3:2
2. **A bead string on a football field has 9 blue and 21 yellow beads. What is the simplified blue:yellow ratio?**
   A) 3:7 B) 3:10 C) 7:3 D) 10:7
3. **A bead string in a recycling club has 3 blue and 6 yellow beads. What is the simplified blue:yellow ratio?**
   A) 3:2 B) 1:2 C) 1:3 D) 2:1
4. **A bead string at a science exhibition has 21 blue and 9 yellow beads. What is the simplified blue:yellow ratio?**
   A) 7:3 B) 10:3 C) 7:10 D) 3:7
5. **A bead string during a reading competition has 6 blue and 21 yellow beads. What is the simplified blue:yellow ratio?**
   A) 2:9 B) 2:7 C) 9:7 D) 7:2

Representing ratios with objects

CHALLENGE · APPLY WHAT YOU KNOW

6 **A bead string for a school fundraiser has 15 blue and 24 yellow beads. What is the simplified blue:yellow ratio?**

A) 13:8 B) 8:5 C) 5:13 D) 5:8

7 **A bead string at a farm has 9 blue and 6 yellow beads. What is the simplified blue:yellow ratio?**

A) 5:2 B) 3:5 C) 2:3 D) 3:2

8 **A bead string during a sports day has 21 blue and 30 yellow beads. What is the simplified blue:yellow ratio?**

A) 7:17 B) 10:7 C) 7:10 D) 17:10

9 **A bead string in a craft workshop has 3 blue and 3 yellow beads. What is the simplified blue:yellow ratio?**

A) 1:1 B) none of these C) 1:2 D) 2:1

10 **A bead string at a school fair has 3 blue and 12 yellow beads. What is the simplified blue:yellow ratio?**

A) 1:5 B) 5:4 C) 1:4 D) 4:1

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 4:5	2. 9:10	3. 5:8
4. 8:5	5. 3:4	6. 3:2
7. 3:7	8. 1:2	9. 7:3
10. 2:7	11. 5:8	12. 3:2
13. 7:10	14. 1:1	15. 1:4

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Making and comparing equivalent ratios

LEARNING GOAL Making and comparing equivalent ratios

MATHS MODEL • RATIO EQUIV



$$3 : 5 = 6 : 10$$

Scale BOTH parts by the same number

LEARN THE IDEA

Equivalent ratios multiply or divide both parts by the same positive factor.

WORKED EXAMPLE 1

Write an equivalent ratio to 7:3 using factor 2.

Answer: 14:6

Multiply both parts by 2 to get 14:6.

WORKED EXAMPLE 2

Write an equivalent ratio to 1:5 using factor 4.

Answer: 4:20

Multiply both parts by 4 to get 4:20.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Making and comparing equivalent ratios

PLAY · LEARN THE STEPS

1. **Write an equivalent ratio to 1:2 using factor 4.**
   A) 8:4 B) 4:8 C) 4:2 D) 5:6
2. **Write an equivalent ratio to 2:1 using factor 3.**
   A) 5:4 B) 6:3 C) 6:1 D) 3:6
3. **Write an equivalent ratio to 8:3 using factor 5.**
   A) 15:40 B) 40:15 C) 13:8 D) 40:3
4. **Write an equivalent ratio to 3:2 using factor 2.**
   A) 4:6 B) 6:2 C) 6:4 D) 5:4
5. **Write an equivalent ratio to 1:1 using factor 3.**
   A) none of these B) 4:4 C) 3:1 D) 3:3

CHALLENGE · DIFFERENT EXAMPLES

1. **A recipe on a football field mixes fruit and water in the ratio 8:5. Which ratio keeps the same taste when quantities are multiplied by 3?**
   A) 15:24 B) 11:8 C) 24:5 D) 24:15
2. **A recipe in a recycling club mixes fruit and water in the ratio 2:5. Which ratio keeps the same taste when quantities are multiplied by 5?**
   A) 7:10 B) 10:5 C) 10:25 D) 25:10
3. **A recipe at a science exhibition mixes fruit and water in the ratio 3:2. Which ratio keeps the same taste when quantities are multiplied by 5?**
   A) 10:15 B) 15:2 C) 15:10 D) 8:7
4. **A recipe during a reading competition mixes fruit and water in the ratio 9:7. Which ratio keeps the same taste when quantities are multiplied by 5?**
   A) 35:45 B) 14:12 C) 45:35 D) 45:7
5. **A recipe for a school fundraiser mixes fruit and water in the ratio 3:8. Which ratio keeps the same taste when quantities are multiplied by 4?**
   A) 32:12 B) 12:32 C) 12:8 D) 7:12

Making and comparing equivalent ratios

CHALLENGE · APPLY WHAT YOU KNOW

6 **A recipe at a farm mixes fruit and water in the ratio 4:3. Which ratio keeps the same taste when quantities are multiplied by 2?**

A) 8:6 B) 6:5 C) 8:3 D) 6:8

7 **A recipe during a sports day mixes fruit and water in the ratio 6:7. Which ratio keeps the same taste when quantities are multiplied by 5?**

A) 11:12 B) 30:7 C) 30:35 D) 35:30

8 **A recipe in a craft workshop mixes fruit and water in the ratio 1:4. Which ratio keeps the same taste when quantities are multiplied by 2?**

A) 8:2 B) 2:8 C) 3:6 D) 2:4

9 **A recipe at a school fair mixes fruit and water in the ratio 7:6. Which ratio keeps the same taste when quantities are multiplied by 3?**

A) 18:21 B) 21:18 C) 10:9 D) 21:6

10 **A recipe in the class garden mixes fruit and water in the ratio 1:3. Which ratio keeps the same taste when quantities are multiplied by 3?**

A) 3:9 B) 4:6 C) 9:3 D) 3:3

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 4:8	2. 6:3	3. 40:15
4. 6:4	5. 3:3	6. 24:15
7. 10:25	8. 15:10	9. 45:35
10. 12:32	11. 8:6	12. 30:35
13. 2:8	14. 21:18	15. 3:9

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Recognising proportional relationships

LEARNING GOAL Recognising proportional relationships

MATHS MODEL • PROPORTION



$$3 : 5 = 6 : 10$$

Scale BOTH parts by the same number

LEARN THE IDEA

A proportion states that two ratios are equal. keep the unit rate constant when scaling.

WORKED EXAMPLE 1

If 6 books cost GH¢30, how much do 30 books cost?

Answer: 150

Multiply by 5: 30 × 5 = 150.

WORKED EXAMPLE 2

If 7 books cost GH¢14, how much do 35 books cost?

Answer: 70

Multiply by 5: 14 × 5 = 70.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Recognising proportional relationships

PLAY · LEARN THE STEPS

1. **If 2 books cost GH₵10, how much do 4 books cost?**
   A) 12 B) 25 C) 10 D) 20
2. **If 7 books cost GH₵35, how much do 49 books cost?**
   A) 35 B) 250 C) 245 D) 42
3. **If 7 books cost GH₵28, how much do 21 books cost?**
   A) 31 B) 28 C) 88 D) 84
4. **If 2 books cost GH₵12, how much do 8 books cost?**
   A) 54 B) 16 C) 12 D) 48
5. **If 3 books cost GH₵18, how much do 12 books cost?**
   A) 18 B) 78 C) 72 D) 22

CHALLENGE · DIFFERENT EXAMPLES

1. **7 exercise books cost GH₵42. At the same unit price in a recycling club, how much do 42 books cost?**
   A) 42 B) 36 C) 258 D) 252
2. **5 exercise books cost GH₵30. At the same unit price at a science exhibition, how much do 10 books cost?**
   A) 60 B) 12 C) 30 D) 66
3. **6 exercise books cost GH₵36. At the same unit price during a reading competition, how much do 42 books cost?**
   A) 42 B) 258 C) 36 D) 252
4. **4 exercise books cost GH₵20. At the same unit price for a school fundraiser, how much do 28 books cost?**
   A) 35 B) 140 C) 145 D) 20
5. **5 exercise books cost GH₵10. At the same unit price at a farm, how much do 25 books cost?**
   A) 50 B) 51 C) 52 D) 10

Recognising proportional relationships

CHALLENGE · APPLY WHAT YOU KNOW

6 **2 exercise books cost GH₵18. At the same unit price during a sports day, how much do 14 books cost?**

A) 135 B) 63 C) 126 D) 18

7 **7 exercise books cost GH₵49. At the same unit price in a craft workshop, how much do 42 books cost?**

A) 49 B) 294 C) 42 D) 301

8 **3 exercise books cost GH₵9. At the same unit price at a school fair, how much do 12 books cost?**

A) 12 B) 36 C) 9 D) 39

9 **6 exercise books cost GH₵36. At the same unit price in the class garden, how much do 36 books cost?**

A) 216 B) 217 C) 222 D) 36

10 **6 exercise books cost GH₵54. At the same unit price at a market stall, how much do 30 books cost?**

A) 54 B) 279 C) 270 D) 45

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 20	2. 245	3. 84
4. 48	5. 72	6. 252
7. 60	8. 252	9. 140
10. 50	11. 126	12. 294
13. 36	14. 216	15. 270

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Solving proportional reasoning problems

LEARNING GOAL Solving proportional reasoning problems

MATHS MODEL • PROPORTION PROBLEM



$$3 : 5 = 6 : 10$$

Scale BOTH parts by the same number

LEARN THE IDEA

First find the unit rate or scaling factor, then apply that factor consistently to the new situation.

WORKED EXAMPLE 1

At a fixed rate of 2 items per 2 people, find the items for 6 people.

Answer: 6

Both parts scale by 3, so 2 × 3 = 6.

WORKED EXAMPLE 2

At a fixed rate of 4 items per 5 people, find the items for 25 people.

Answer: 20

Both parts scale by 5, so 4 × 5 = 20.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Solving proportional reasoning problems

PLAY · LEARN THE STEPS

1. **At a fixed rate of 5 items per 3 people, find the items for 9 people.**
   A) 5 B) 16 C) 14 D) 15
2. **At a fixed rate of 3 items per 8 people, find the items for 24 people.**
   A) 3 B) 9 C) 10 D) 27
3. **At a fixed rate of 2 items per 8 people, find the items for 48 people.**
   A) 13 B) 50 C) 12 D) 2
4. **At a fixed rate of 2 items per 2 people, find the items for 6 people.**
   A) 7 B) 2 C) 8 D) 6
5. **At a fixed rate of 4 items per 5 people, find the items for 10 people.**
   A) 8 B) 14 C) 9 D) 4

CHALLENGE · DIFFERENT EXAMPLES

1. **A class at a science exhibition uses 3 bottles for 3 pupils. At the same rate, how many bottles are needed for 6 pupils?**
   A) 6 B) 7 C) 9 D) 3
2. **A class during a reading competition uses 5 bottles for 6 pupils. At the same rate, how many bottles are needed for 30 pupils?**
   A) 25 B) 5 C) 35 D) 26
3. **A class for a school fundraiser uses 6 bottles for 7 pupils. At the same rate, how many bottles are needed for 14 pupils?**
   A) 20 B) 6 C) 12 D) 13
4. **A class at a farm uses 3 bottles for 8 pupils. At the same rate, how many bottles are needed for 32 pupils?**
   A) 12 B) 3 C) 13 D) 35
5. **A class during a sports day uses 7 bottles for 3 pupils. At the same rate, how many bottles are needed for 9 pupils?**
   A) 22 B) 7 C) 16 D) 21

Solving proportional reasoning problems

CHALLENGE · APPLY WHAT YOU KNOW

6 **A class in a craft workshop uses 4 bottles for 9 pupils. At the same rate, how many bottles are needed for 18 pupils?**

A) 8 B) 9 C) 22 D) 4

7 **A class at a school fair uses 4 bottles for 8 pupils. At the same rate, how many bottles are needed for 16 pupils?**

A) 8 B) 9 C) 20 D) 4

8 **A class in the class garden uses 4 bottles for 9 pupils. At the same rate, how many bottles are needed for 45 pupils?**

A) 21 B) 4 C) 49 D) 20

9 **A class at a market stall uses 7 bottles for 4 pupils. At the same rate, how many bottles are needed for 12 pupils?**

A) 21 B) 22 C) 7 D) 19

10 **A class during a library project uses 5 bottles for 8 pupils. At the same rate, how many bottles are needed for 32 pupils?**

A) 21 B) 37 C) 20 D) 5

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 15	2. 9	3. 12
4. 6	5. 8	6. 6
7. 25	8. 12	9. 12
10. 21	11. 8	12. 8
13. 20	14. 21	15. 20

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Visual number patterns and predictions

LEARNING GOAL Visual number patterns and predictions

MATHS MODEL • VISUAL PATTERNS



Find a rule • Substitute • Explain

LEARN THE IDEA

Draw or count repeated tile groups; use the difference to predict the number in later figures.

WORKED EXAMPLE 1

Figure counts: 1, 3, 5, ... What is figure 4?

Answer: 7

Add 2 each time: $1+(4-1)\times 2=7$.

WORKED EXAMPLE 2

Figure counts: 4, 10, 16, ... What is figure 7?

Answer: 40

Add 6 each time: $4+(7-1)\times 6=40$.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Visual number patterns and predictions

PLAY · LEARN THE STEPS

1. **Figure counts: 4, 7, 10, ... What is figure 7?**
   A) 22 B) 25 C) 23 D) 19
2. **Figure counts: 1, 8, 15, ... What is figure 8?**
   A) 50 B) 43 C) 57 D) 51
3. **Figure counts: 4, 8, 12, ... What is figure 6?**
   A) 28 B) 20 C) 25 D) 24
4. **Figure counts: 3, 8, 13, ... What is figure 5?**
   A) 28 B) 18 C) 23 D) 24
5. **Figure counts: 3, 5, 7, ... What is figure 4?**
   A) 11 B) 7 C) 9 D) 10

CHALLENGE · DIFFERENT EXAMPLES

1. **A tile design during a reading competition has 4, 9, 14, ... tiles in successive figures. How many tiles are in figure 9?**
   A) 45 B) 44 C) 49 D) 39
2. **A tile design for a school fundraiser has 6, 10, 14, ... tiles in successive figures. How many tiles are in figure 7?**
   A) 26 B) 31 C) 34 D) 30
3. **A tile design at a farm has 5, 7, 9, ... tiles in successive figures. How many tiles are in figure 8?**
   A) 21 B) 19 C) 17 D) 20
4. **A tile design during a sports day has 2, 9, 16, ... tiles in successive figures. How many tiles are in figure 8?**
   A) 52 B) 51 C) 44 D) 58
5. **A tile design in a craft workshop has 2, 8, 14, ... tiles in successive figures. How many tiles are in figure 7?**
   A) 32 B) 44 C) 38 D) 39

Visual number patterns and predictions

CHALLENGE · APPLY WHAT YOU KNOW

6 **A tile design at a school fair has 6, 10, 14, ... tiles in successive figures. How many tiles are in figure 4?**

A) 22 B) 14 C) 18 D) 19

7 **A tile design in the class garden has 6, 8, 10, ... tiles in successive figures. How many tiles are in figure 8?**

A) 21 B) 20 C) 18 D) 22

8 **A tile design at a market stall has 2, 6, 10, ... tiles in successive figures. How many tiles are in figure 8?**

A) 30 B) 26 C) 31 D) 34

9 **A tile design during a library project has 3, 8, 13, ... tiles in successive figures. How many tiles are in figure 8?**

A) 39 B) 43 C) 38 D) 33

10 **A tile design on a football field has 5, 8, 11, ... tiles in successive figures. How many tiles are in figure 8?**

A) 29 B) 27 C) 26 D) 23

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 22	2. 50	3. 24
4. 23	5. 9	6. 44
7. 30	8. 19	9. 51
10. 38	11. 18	12. 20
13. 30	14. 38	15. 26

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Writing algebraic pattern rules

LEARNING GOAL Writing algebraic pattern rules

MATHS MODEL • PATTERN RULE



Describe the rule, then explain why it works.

LEARN THE IDEA

A pattern rule expresses the starting amount and change using a letter such as n.

WORKED EXAMPLE 1

A rule starts at 25 and increases by 4 per step. Which expression describes it?

Answer: 25 + 4n

Initial value 25, then add 4 for each step n.

WORKED EXAMPLE 2

A rule starts at 16 and increases by 10 per step. Which expression describes it?

Answer: 16 + 10n

Initial value 16, then add 10 for each step n.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Writing algebraic pattern rules

PLAY · LEARN THE STEPS

1. **A rule starts at 12 and increases by 10 per step. Which expression describes it?**
   A) 12n + 10 B) 12 + n + 10 C) 12 - 10n D) 12 + 10n
2. **A rule starts at 1 and increases by 9 per step. Which expression describes it?**
   A) 1n + 9 B) 1 - 9n C) 1 + n + 9 D) 1 + 9n
3. **A rule starts at 9 and increases by 8 per step. Which expression describes it?**
   A) 9 + 8n B) 9 - 8n C) 9 + n + 8 D) 9n + 8
4. **A rule starts at 18 and increases by 6 per step. Which expression describes it?**
   A) 18 - 6n B) 18n + 6 C) 18 + 6n D) 18 + n + 6
5. **A rule starts at 4 and increases by 5 per step. Which expression describes it?**
   A) 4 + 5n B) 4 - 5n C) 4 + n + 5 D) 4n + 5

CHALLENGE · DIFFERENT EXAMPLES

1. **A price table in a recycling club starts at 22 and adds 2 for each new step. Which expression gives the value at step n?**
   A) 22 - 2n B) 22n + 2 C) 22 + 2n D) 22 + n + 2
2. **A price table at a science exhibition starts at 1 and adds 5 for each new step. Which expression gives the value at step n?**
   A) 1 - 5n B) 1 + n + 5 C) 1n + 5 D) 1 + 5n
3. **A price table during a reading competition starts at 1 and adds 2 for each new step. Which expression gives the value at step n?**
   A) 1 + n + 2 B) 1n + 2 C) 1 + 2n D) 1 - 2n
4. **A price table for a school fundraiser starts at 9 and adds 10 for each new step. Which expression gives the value at step n?**
   A) 9 + 10n B) 9 + n + 10 C) 9 - 10n D) 9n + 10
5. **A price table at a farm starts at 22 and adds 6 for each new step. Which expression gives the value at step n?**
   A) 22 + 6n B) 22 + n + 6 C) 22n + 6 D) 22 - 6n

Writing algebraic pattern rules

CHALLENGE · APPLY WHAT YOU KNOW

6 **A price table during a sports day starts at 3 and adds 10 for each new step. Which expression gives the value at step n?**

A) 3 - 10n B) 3n + 10 C) 3 + 10n D) 3 + n + 10

7 **A price table in a craft workshop starts at 12 and adds 2 for each new step. Which expression gives the value at step n?**

A) 12n + 2 B) 12 - 2n C) 12 + 2n D) 12 + n + 2

8 **A price table at a school fair starts at 17 and adds 4 for each new step. Which expression gives the value at step n?**

A) 17n + 4 B) 17 - 4n C) 17 + n + 4 D) 17 + 4n

9 **A price table in the class garden starts at 12 and adds 6 for each new step. Which expression gives the value at step n?**

A) 12 + n + 6 B) 12 - 6n C) 12n + 6 D) 12 + 6n

10 **A price table at a market stall starts at 17 and adds 10 for each new step. Which expression gives the value at step n?**

A) 17 + n + 10 B) 17 - 10n C) 17 + 10n D) 17n + 10

ANSWER CHECK · LOOK ONLY AFTER TRYING

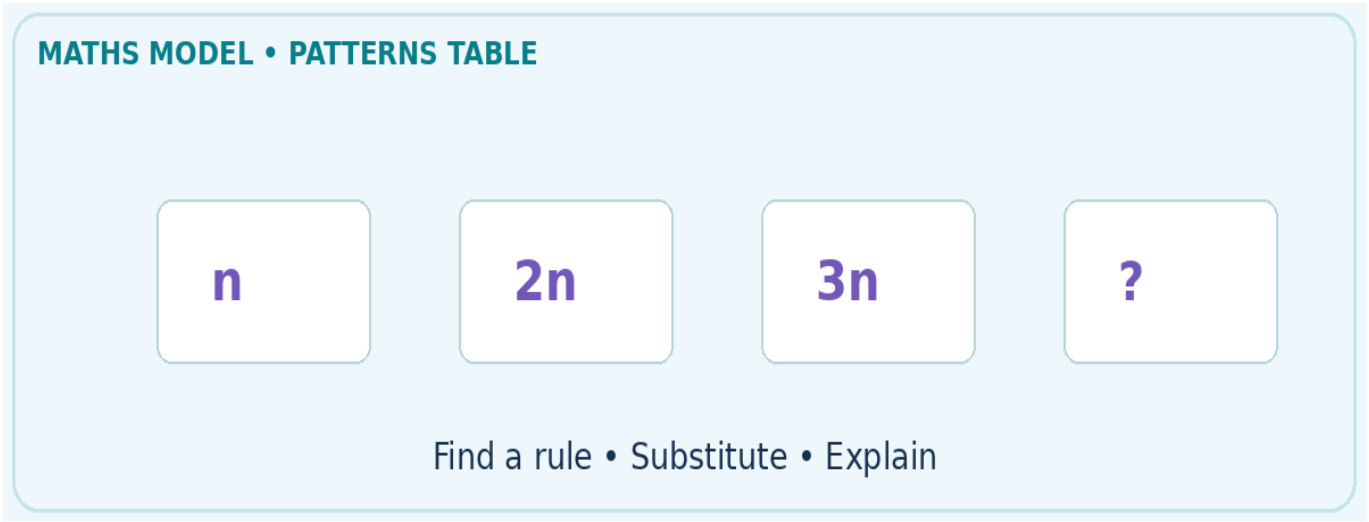
1. 12 + 10n	2. 1 + 9n	3. 9 + 8n
4. 18 + 6n	5. 4 + 5n	6. 22 + 2n
7. 1 + 5n	8. 1 + 2n	9. 9 + 10n
10. 22 + 6n	11. 3 + 10n	12. 12 + 2n
13. 17 + 4n	14. 12 + 6n	15. 17 + 10n

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Describing number tables with expressions

LEARNING GOAL Describing number tables with expressions



LEARN THE IDEA

A table links input and output. describe the pattern as an algebraic rule and test it with two rows.

WORKED EXAMPLE 1

A table follows y = 6n + 5. Calculate y for n=4.

Answer: 29

6×4+5 = 29.

WORKED EXAMPLE 2

A table follows y = 11n + 3. Calculate y for n=7.

Answer: 80

11×7+3 = 80.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Describing number tables with expressions

PLAY · LEARN THE STEPS

1. **A table follows y = 6n + 4. Calculate y for n=5.**
   A) 34 B) 30 C) 40 D) 35
2. **A table follows y = 9n + 5. Calculate y for n=9.**
   A) 95 B) 86 C) 81 D) 87
3. **A table follows y = 5n + 1. Calculate y for n=5.**
   A) 27 B) 26 C) 31 D) 25
4. **A table follows y = 7n + 5. Calculate y for n=4.**
   A) 40 B) 34 C) 33 D) 28
5. **A table follows y = 7n + 3. Calculate y for n=5.**
   A) 35 B) 45 C) 39 D) 38

CHALLENGE · DIFFERENT EXAMPLES

1. **A fare chart at a farm follows output = 8 × input + 0. What output matches input 5?**
   A) 48 B) 41 C) 40 D) 42
2. **A fare chart during a sports day follows output = 3 × input + 5. What output matches input 9?**
   A) 33 B) 32 C) 27 D) 35
3. **A fare chart in a craft workshop follows output = 6 × input + 1. What output matches input 6?**
   A) 38 B) 36 C) 37 D) 43
4. **A fare chart at a school fair follows output = 4 × input + 6. What output matches input 10?**
   A) 46 B) 40 C) 50 D) 47
5. **A fare chart in the class garden follows output = 5 × input + 3. What output matches input 7?**
   A) 39 B) 35 C) 43 D) 38

Describing number tables with expressions

CHALLENGE · APPLY WHAT YOU KNOW

6 **A fare chart at a market stall follows output = 10 × input + 0. What output matches input 5?**

A) 60 B) 50 C) 52 D) 51

7 **A fare chart during a library project follows output = 6 × input + 5. What output matches input 8?**

A) 53 B) 59 C) 48 D) 54

8 **A fare chart on a football field follows output = 4 × input + 0. What output matches input 12?**

A) 52 B) 50 C) 49 D) 48

9 **A fare chart in a recycling club follows output = 9 × input + 6. What output matches input 9?**

A) 81 B) 96 C) 87 D) 88

10 **A fare chart at a science exhibition follows output = 3 × input + 2. What output matches input 4?**

A) 17 B) 15 C) 14 D) 12

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 34	2. 86	3. 26
4. 33	5. 38	6. 40
7. 32	8. 37	9. 46
10. 38	11. 50	12. 53
13. 48	14. 87	15. 14

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Writing algebraic expressions

LEARNING GOAL Writing algebraic expressions

MATHS MODEL • ALGEBRA WORDS



Find a rule • Substitute • Explain

LEARN THE IDEA

Translate phrases into operations. “four times n plus three” means 4n + 3.

WORKED EXAMPLE 1

Write an expression for 7 times x, then add 9.

Answer: 7x + 9

Multiply the variable by 7, then add the fixed 9.

WORKED EXAMPLE 2

Write an expression for 9 times x, then add 5.

Answer: 9x + 5

Multiply the variable by 9, then add the fixed 5.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Writing algebraic expressions

PLAY · LEARN THE STEPS

1. **Write an expression for 3 times x, then add 6.**
   A) 9x B) 3x − 6 C) 3x + 6 D) 3+x+6
2. **Write an expression for 6 times x, then add 4.**
   A) 6+x+4 B) 6x − 4 C) 10x D) 6x + 4
3. **Write an expression for 7 times x, then add 2.**
   A) 7x − 2 B) 7+x+2 C) 9x D) 7x + 2
4. **Write an expression for 3 times x, then add 3.**
   A) 3x − 3 B) 6x C) 3x + 3 D) 3+x+3
5. **Write an expression for 2 times x, then add 11.**
   A) 2x + 11 B) 13x C) 2+x+11 D) 2x − 11

CHALLENGE · DIFFERENT EXAMPLES

1. **A school shop during a sports day charges GH¢2 per pen plus GH¢3 delivery. Which expression gives the total for x pens?**
   A) 5x B) 2x − 3 C) 2+x+3 D) 2x + 3
2. **A school shop in a craft workshop charges GH¢5 per pen plus GH¢12 delivery. Which expression gives the total for x pens?**
   A) 5x − 12 B) 17x C) 5x + 12 D) 5+x+12
3. **A school shop at a school fair charges GH¢7 per pen plus GH¢10 delivery. Which expression gives the total for x pens?**
   A) 7x − 10 B) 7x + 10 C) 7+x+10 D) 17x
4. **A school shop in the class garden charges GH¢7 per pen plus GH¢5 delivery. Which expression gives the total for x pens?**
   A) 7+x+5 B) 12x C) 7x + 5 D) 7x − 5
5. **A school shop at a market stall charges GH¢7 per pen plus GH¢2 delivery. Which expression gives the total for x pens?**
   A) 7+x+2 B) 7x + 2 C) 9x D) 7x − 2

Writing algebraic expressions

CHALLENGE · APPLY WHAT YOU KNOW

6 **A school shop during a library project charges GH¢9 per pen plus GH¢2 delivery. Which expression gives the total for x pens?**

A) 11x B) 9x − 2 C) 9+x+2 D) 9x + 2

7 **A school shop on a football field charges GH¢9 per pen plus GH¢11 delivery. Which expression gives the total for x pens?**

A) 9x + 11 B) 20x C) 9+x+11 D) 9x − 11

8 **A school shop in a recycling club charges GH¢5 per pen plus GH¢10 delivery. Which expression gives the total for x pens?**

A) 5+x+10 B) 5x + 10 C) 15x D) 5x − 10

9 **A school shop at a science exhibition charges GH¢5 per pen plus GH¢9 delivery. Which expression gives the total for x pens?**

A) 14x B) 5x + 9 C) 5x − 9 D) 5+x+9

10 **A school shop during a reading competition charges GH¢6 per pen plus GH¢3 delivery. Which expression gives the total for x pens?**

A) 6x + 3 B) 6+x+3 C) 9x D) 6x − 3

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 3x + 6	2. 6x + 4	3. 7x + 2
4. 3x + 3	5. 2x + 11	6. 2x + 3
7. 5x + 12	8. 7x + 10	9. 7x + 5
10. 7x + 2	11. 9x + 2	12. 9x + 11
13. 5x + 10	14. 5x + 9	15. 6x + 3

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Combining like terms

LEARNING GOAL Combining like terms

MATHS MODEL • LIKE TERMS



Find a rule • Substitute • Explain

LEARN THE IDEA

Only add coefficients of like terms. constants combine with constants, and x-terms with x-terms.

WORKED EXAMPLE 1

Simplify 9x + 5x + 4 − 5.

Answer: 14x − 1

Combine x-terms (9+5)x and constants 4−5.

WORKED EXAMPLE 2

Simplify 8x + 2x + 8 − 2.

Answer: 10x + 6

Combine x-terms (8+2)x and constants 8−2.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Combining like terms

PLAY · LEARN THE STEPS

1. **Simplify 5x + 5x + 7 − 8.**
   A) 9x B) 10x − 1 C) 11x + 7 D) 25x + -1
2. **Simplify 3x + 5x + 2 − 4.**
   A) 8x − 2 B) 9x + 2 C) 15x + -2 D) 6x
3. **Simplify 9x + 3x + 6 − 7.**
   A) 12x − 1 B) 11x C) 27x + -1 D) 13x + 6
4. **Simplify 6x + 5x + 7 − 8.**
   A) 10x B) 30x + -1 C) 11x − 1 D) 12x + 7
5. **Simplify 8x + 6x + 6 − 3.**
   A) 17x B) 48x + 3 C) 14x + 3 D) 15x + 6

CHALLENGE · DIFFERENT EXAMPLES

1. **A learner in a craft workshop combines 8x + 6 + 4x − 7. Which simplified expression is correct?**
   A) 12x − 1 B) 32x + -1 C) 11x D) 13x + 6
2. **A learner at a school fair combines 5x + 3 + 3x − 4. Which simplified expression is correct?**
   A) 8x − 1 B) 7x C) 9x + 3 D) 15x + -1
3. **A learner in the class garden combines 9x + 6 + 5x − 2. Which simplified expression is correct?**
   A) 14x + 4 B) 18x C) 15x + 6 D) 45x + 4
4. **A learner at a market stall combines 2x + 5 + 6x − 6. Which simplified expression is correct?**
   A) 8x − 1 B) 7x C) 12x + -1 D) 9x + 5
5. **A learner during a library project combines 6x + 3 + 4x − 6. Which simplified expression is correct?**
   A) 24x + -3 B) 7x C) 11x + 3 D) 10x − 3

Combining like terms

CHALLENGE · APPLY WHAT YOU KNOW

6 **A learner on a football field combines 5x + 3 + 9x − 1. Which simplified expression is correct?**

A) 15x + 3 B) 16x C) 45x + 2 D) 14x + 2

7 **A learner in a recycling club combines 5x + 6 + 4x − 2. Which simplified expression is correct?**

A) 10x + 6 B) 9x + 4 C) 20x + 4 D) 13x

8 **A learner at a science exhibition combines 6x + 2 + 3x − 4. Which simplified expression is correct?**

A) 9x − 2 B) 18x + -2 C) 10x + 2 D) 7x

9 **A learner during a reading competition combines 8x + 2 + 8x − 7. Which simplified expression is correct?**

A) 64x + -5 B) 16x − 5 C) 17x + 2 D) 11x

10 **A learner for a school fundraiser combines 8x + 7 + 2x − 1. Which simplified expression is correct?**

A) 16x + 6 B) 11x + 7 C) 10x + 6 D) 16x

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 10x − 1	2. 8x − 2	3. 12x − 1
4. 11x − 1	5. 14x + 3	6. 12x − 1
7. 8x − 1	8. 14x + 4	9. 8x − 1
10. 10x − 3	11. 14x + 2	12. 9x + 4
13. 9x − 2	14. 16x − 5	15. 10x + 6

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Substituting values in expressions

LEARNING GOAL Substituting values in expressions

MATHS MODEL • SUBSTITUTE



Find a rule • Substitute • Explain

LEARN THE IDEA

Replace the letter with the given number, then use the order of operations.

WORKED EXAMPLE 1

Evaluate 5n + 7 if n=8.

Answer: 47

Substitute n=8 to obtain 5×8+7=47.

WORKED EXAMPLE 2

Evaluate 2n + 2 if n=5.

Answer: 12

Substitute n=5 to obtain 2×5+2=12.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Substituting values in expressions

PLAY · LEARN THE STEPS

1. **Evaluate 2n + 8 if n=8.**
   A) 26 B) 16 C) 24 D) 18
2. **Evaluate 8n + 6 if n=7.**
   A) 70 B) 21 C) 56 D) 62
3. **Evaluate 3n + 6 if n=2.**
   A) 11 B) 15 C) 12 D) 6
4. **Evaluate 4n + 2 if n=2.**
   A) 8 B) 14 C) 11 D) 10
5. **Evaluate 4n + 8 if n=7.**
   A) 19 B) 36 C) 40 D) 28

CHALLENGE · DIFFERENT EXAMPLES

1. **A display at a school fair uses rule 9n + 8. What is the display when n=2?**
   A) 19 B) 18 C) 35 D) 26
2. **A display in the class garden uses rule 5n + 6. What is the display when n=9?**
   A) 51 B) 45 C) 56 D) 20
3. **A display at a market stall uses rule 2n + 6. What is the display when n=2?**
   A) 11 B) 4 C) 10 D) 12
4. **A display during a library project uses rule 3n + 3. What is the display when n=2?**
   A) 6 B) 9 C) 8 D) 12
5. **A display on a football field uses rule 6n + 7. What is the display when n=5?**
   A) 37 B) 43 C) 30 D) 18

Substituting values in expressions

CHALLENGE · APPLY WHAT YOU KNOW

6. **A display in a recycling club uses rule 4n + 4. What is the display when n=2?**
   A) 10 B) 12 C) 16 D) 8

7. **A display at a science exhibition uses rule 3n + 8. What is the display when n=9?**
   A) 27 B) 20 C) 38 D) 35

8. **A display during a reading competition uses rule 5n + 6. What is the display when n=6?**
   A) 41 B) 30 C) 36 D) 17

9. **A display for a school fundraiser uses rule 6n + 3. What is the display when n=2?**
   A) 21 B) 12 C) 11 D) 15

10. **A display at a farm uses rule 7n + 5. What is the display when n=4?**
    A) 33 B) 28 C) 40 D) 16

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 24	2. 62	3. 12
4. 10	5. 36	6. 26
7. 51	8. 10	9. 9
10. 37	11. 12	12. 35
13. 36	14. 15	15. 33

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Solving one-step equations

LEARNING GOAL Solving one-step equations

MATHS MODEL • EQUATIONS



Describe the rule, then explain why it works.

LEARN THE IDEA

An equation says two quantities are equal. use inverse operations to solve for the unknown.

WORKED EXAMPLE 1

Solve x + 4 = 30.

Answer: 26

x = 30 − 4 = 26.

WORKED EXAMPLE 2

Solve x + 8 = 14.

Answer: 6

x = 14 − 8 = 6.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Solving one-step equations

PLAY · LEARN THE STEPS

1. **Solve x + 11 = 33.**
   A) 24 B) 11 C) 33 D) 22
2. **Solve x + 18 = 38.**
   A) 20 B) 18 C) 22 D) 38
3. **Solve x + 16 = 31.**
   A) 16 B) 15 C) 17 D) 31
4. **Solve x + 11 = 37.**
   A) 11 B) 37 C) 26 D) 28
5. **Solve x + 18 = 35.**
   A) 19 B) 35 C) 18 D) 17

CHALLENGE · DIFFERENT EXAMPLES

1. **A box during a sports day has 11 stickers plus x more; altogether there are 34. What is x?**
   A) 34 B) 25 C) 23 D) 11
2. **A box in a craft workshop has 23 stickers plus x more; altogether there are 34. What is x?**
   A) 11 B) 23 C) 34 D) 13
3. **A box at a school fair has 24 stickers plus x more; altogether there are 30. What is x?**
   A) 6 B) 8 C) 24 D) 30
4. **A box in the class garden has 5 stickers plus x more; altogether there are 32. What is x?**
   A) 27 B) 32 C) 5 D) 29
5. **A box at a market stall has 7 stickers plus x more; altogether there are 32. What is x?**
   A) 27 B) 7 C) 25 D) 32

Solving one-step equations

CHALLENGE · APPLY WHAT YOU KNOW

6. **A box during a library project has 11 stickers plus x more; altogether there are 22. What is x?**
   A) 22 B) 13 C) 12 D) 11

7. **A box on a football field has 21 stickers plus x more; altogether there are 30. What is x?**
   A) 30 B) 21 C) 9 D) 11

8. **A box in a recycling club has 5 stickers plus x more; altogether there are 20. What is x?**
   A) 15 B) 20 C) 17 D) 5

9. **A box at a science exhibition has 21 stickers plus x more; altogether there are 50. What is x?**
   A) 29 B) 50 C) 31 D) 21

10. **A box during a reading competition has 12 stickers plus x more; altogether there are 34. What is x?**
    A) 12 B) 22 C) 34 D) 24

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 22	2. 20	3. 15
4. 26	5. 17	6. 23
7. 11	8. 6	9. 27
10. 25	11. 11	12. 9
13. 15	14. 29	15. 22

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Creating real stories for equations

LEARNING GOAL Creating real stories for equations

MATHS MODEL • EQUATION STORY



Find a rule • Substitute • Explain

LEARN THE IDEA

An equation tells a story about equal quantities. check that the operation and unknown match the story.

WORKED EXAMPLE 1

Find x in $x - 9 = 4$.

Answer: 13

Add 9 to both sides; x = 13. You can also tell a story that matches the equation.

WORKED EXAMPLE 2

Find x in $x - 7 = 4$.

Answer: 11

Add 7 to both sides; x = 11. You can also tell a story that matches the equation.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Creating real stories for equations

PLAY · LEARN THE STEPS

1. **Find x in x − 3 = 11.**
   A) 11 B) 16 C) 8 D) 14
2. **Find x in x − 6 = 8.**
   A) 8 B) 2 C) 14 D) 16
3. **Find x in x − 15 = 7.**
   A) 24 B) -8 C) 7 D) 22
4. **Find x in x − 13 = 8.**
   A) 23 B) -5 C) 21 D) 8
5. **Find x in x − 8 = 8.**
   A) 16 B) 8 C) 0 D) 18

CHALLENGE · DIFFERENT EXAMPLES

1. **A volunteer at a market stall had an unknown number x of flyers, gave away 8 and had 16 remaining. Which equation models the story?**
   A) x + 8 = 16 B) x − 8 = 16 C) 8 − x = 16 D) x − 16 = 8
2. **A box of unknown pencils during a library project lost 12 pencils; 20 remain. How many were in the box originally?**
   A) 8 B) 32 C) 20 D) 34
3. **A class on a football field had x bottles and received 15 more to make 33. Which equation represents the number it began with?**
   A) x + 33 = 15 B) x − 15 = 33 C) x + 15 = 33 D) 15 − x = 33
4. **An equation in a recycling club is x − 13 = 10. Which story represents it?**
   A) A learner started with x counters, removed 13, and had 10 left. B) A learner had 13 counters and gained 10 to reach x. C) A learner started with 10 counters and removed 13 to reach x. D) A learner started with x counters, added 13, and had 10.
5. **A volunteer at a science exhibition had an unknown number x of flyers, gave away 4 and had 22 remaining. Which equation models the story?**
   A) x − 22 = 4 B) 4 − x = 22 C) x + 4 = 22 D) x − 4 = 22

Creating real stories for equations

CHALLENGE · APPLY WHAT YOU KNOW

6 **A box of unknown pencils during a reading competition lost 3 pencils; 9 remain. How many were in the box originally?**

A) 6 B) 9 C) 14 D) 12

7 **A class for a school fundraiser had x bottles and received 4 more to make 22. Which equation represents the number it began with?**

A) x − 4 = 22 B) x + 4 = 22 C) 4 − x = 22 D) x + 22 = 4

8 **An equation at a farm is x − 4 = 17. Which story represents it?**

A) A learner had 4 counters and gained 17 to reach x. B) A learner started with 17 counters and removed 4 to reach x. C) A learner started with x counters, removed 4, and had 17 left. D) A learner started with x counters, added 4, and had 17.

9 **A volunteer during a sports day had an unknown number x of flyers, gave away 12 and had 13 remaining. Which equation models the story?**

A) 12 − x = 13 B) x + 12 = 13 C) x − 12 = 13 D) x − 13 = 12

10 **A box of unknown pencils in a craft workshop lost 12 pencils; 25 remain. How many were in the box originally?**

A) 39 B) 25 C) 13 D) 37

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 14	2. 14	3. 22
4. 21	5. 16	6. x − 8 = 16
7. 32	8. x + 15 = 33	9. A learner started with x counters, removed 13, and had 10 left.
10. x − 4 = 22	11. 12	12. x + 4 = 22
13. A learner started with x …	14. x − 12 = 13	15. 37

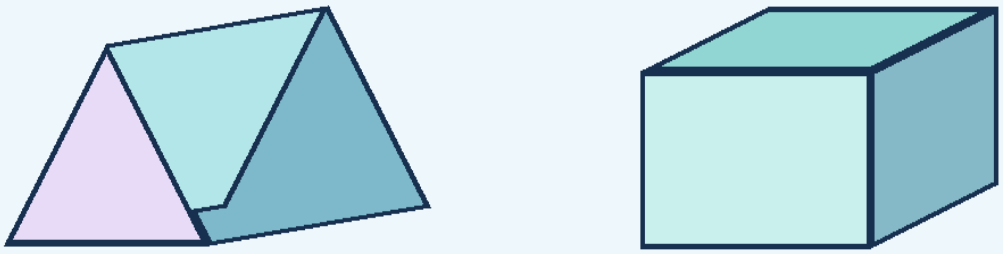
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Recognising rectangular and triangular prisms

LEARNING GOAL Recognising rectangular and triangular prisms

MATHS MODEL • PRISMS



Triangular prism • Rectangular prism

LEARN THE IDEA

A prism has a uniform cross-section along its length; a rectangular prism has 6 faces and a triangular prism has 5.

WORKED EXAMPLE 1

How many flat faces does a rectangular prism have?

Answer: 6

A rectangular prism has 6 faces, 12 edges, 8 vertices and two matching end faces.

WORKED EXAMPLE 2

How many straight edges does a triangular prism have?

Answer: 9

A triangular prism has 5 faces, 9 edges, 6 vertices and two matching end faces.

TRY THIS IN REAL LIFE

Use folded paper, a ruler or safe classroom objects to model each shape or direction.

Recognising rectangular and triangular prisms

PLAY · LEARN THE STEPS

1. **How many flat faces does a rectangular prism have?**
   A) 8 B) 6 C) 7 D) 12
2. **How many straight edges does a triangular prism have?**
   A) 10 B) 6 C) 9 D) 5
3. **How many corners (vertices) does a rectangular prism have?**
   A) 8 B) 6 C) 12 D) 9
4. **How many matching end faces does a triangular prism have?**
   A) 6 B) 2 C) 5 D) 9
5. **How many sides on each end face does a rectangular prism have?**
   A) 4 B) 12 C) 8 D) 6

CHALLENGE · DIFFERENT EXAMPLES

1. **A triangular prism in a display during a library project has two matching parallel ends. How many flat faces does this solid have?**
   A) 5 B) 6 C) 7 D) 9
2. **A rectangular prism in a display on a football field has two matching parallel ends. How many straight edges does this solid have?**
   A) 12 B) 8 C) 6 D) 13
3. **A triangular prism in a display in a recycling club has two matching parallel ends. How many corners (vertices) does this solid have?**
   A) 6 B) 9 C) 7 D) 5
4. **A rectangular prism in a display at a science exhibition has two matching parallel ends. How many matching end faces does this solid have?**
   A) 6 B) 8 C) 12 D) 2
5. **A triangular prism in a display during a reading competition has two matching parallel ends. How many sides on each end face does this solid have?**
   A) 3 B) 9 C) 6 D) 5

Recognising rectangular and triangular prisms

CHALLENGE · APPLY WHAT YOU KNOW

6 **A rectangular prism in a display for a school fundraiser has two matching parallel ends. How many flat faces does this solid have?**

A) 7 B) 8 C) 6 D) 12

7 **A triangular prism in a display at a farm has two matching parallel ends. How many straight edges does this solid have?**

A) 6 B) 10 C) 9 D) 5

8 **A rectangular prism in a display during a sports day has two matching parallel ends. How many corners (vertices) does this solid have?**

A) 6 B) 8 C) 9 D) 12

9 **A triangular prism in a display in a craft workshop has two matching parallel ends. How many matching end faces does this solid have?**

A) 6 B) 2 C) 5 D) 9

10 **A rectangular prism in a display at a school fair has two matching parallel ends. How many sides on each end face does this solid have?**

A) 6 B) 8 C) 12 D) 4

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 6	2. 9	3. 8
4. 2	5. 4	6. 5
7. 12	8. 6	9. 2
10. 3	11. 6	12. 9
13. 8	14. 2	15. 4

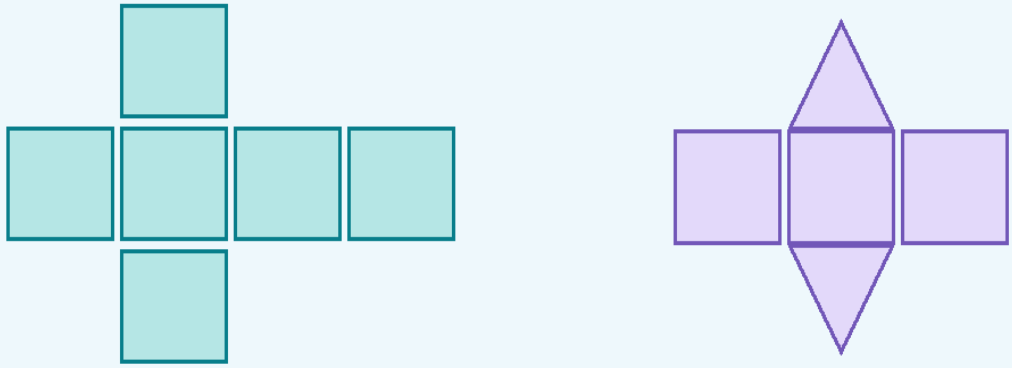
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Nets of prisms

LEARNING GOAL Nets of prisms

MATHS MODEL • NETS



Two connected nets, each ready to fold into a prism

LEARN THE IDEA

A net is a two-dimensional arrangement of faces that folds into a solid. count and position all faces.

WORKED EXAMPLE 1

For the net of a rectangular prism, how many flat faces must be represented?

Answer: 6

A rectangular prism has 6 faces, 12 edges, 8 vertices and two matching end faces.

WORKED EXAMPLE 2

For the net of a triangular prism, how many straight edges must be represented?

Answer: 9

A triangular prism has 5 faces, 9 edges, 6 vertices and two matching end faces.

TRY THIS IN REAL LIFE

Use folded paper, a ruler or safe classroom objects to model each shape or direction.

Nets of prisms

PLAY · LEARN THE STEPS

1. **For the net of a rectangular prism, how many flat faces must be represented?**
   A) 12 B) 6 C) 8 D) 7
2. **For the net of a triangular prism, how many straight edges must be represented?**
   A) 9 B) 10 C) 6 D) 5
3. **For the net of a rectangular prism, how many corners (vertices) must be represented?**
   A) 12 B) 9 C) 8 D) 6
4. **For the net of a triangular prism, how many matching end faces must be represented?**
   A) 6 B) 5 C) 2 D) 9
5. **For the net of a rectangular prism, how many sides on each end face must be represented?**
   A) 12 B) 4 C) 8 D) 6

CHALLENGE · DIFFERENT EXAMPLES

1. **A child on a football field draws a flat net to fold a triangular prism. How many flat faces will the finished shape have?**
   A) 7 B) 9 C) 5 D) 6
2. **A child in a recycling club draws a flat net to fold a rectangular prism. How many straight edges will the finished shape have?**
   A) 12 B) 6 C) 13 D) 8
3. **A child at a science exhibition draws a flat net to fold a triangular prism. How many corners (vertices) will the finished shape have?**
   A) 5 B) 7 C) 9 D) 6
4. **A child during a reading competition draws a flat net to fold a rectangular prism. How many matching end faces will the finished shape have?**
   A) 12 B) 8 C) 6 D) 2
5. **A child for a school fundraiser draws a flat net to fold a triangular prism. How many sides on each end face will the finished shape have?**
   A) 3 B) 9 C) 5 D) 6

Nets of prisms

CHALLENGE · APPLY WHAT YOU KNOW

6 **A child at a farm draws a flat net to fold a rectangular prism. How many flat faces will the finished shape have?**

A) 8 B) 12 C) 7 D) 6

7 **A child during a sports day draws a flat net to fold a triangular prism. How many straight edges will the finished shape have?**

A) 5 B) 10 C) 6 D) 9

8 **A child in a craft workshop draws a flat net to fold a rectangular prism. How many corners (vertices) will the finished shape have?**

A) 8 B) 12 C) 6 D) 9

9 **A child at a school fair draws a flat net to fold a triangular prism. How many matching end faces will the finished shape have?**

A) 6 B) 9 C) 2 D) 5

10 **A child in the class garden draws a flat net to fold a rectangular prism. How many sides on each end face will the finished shape have?**

A) 12 B) 8 C) 4 D) 6

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 6	2. 9	3. 8
4. 2	5. 4	6. 5
7. 12	8. 6	9. 2
10. 3	11. 6	12. 9
13. 8	14. 2	15. 4

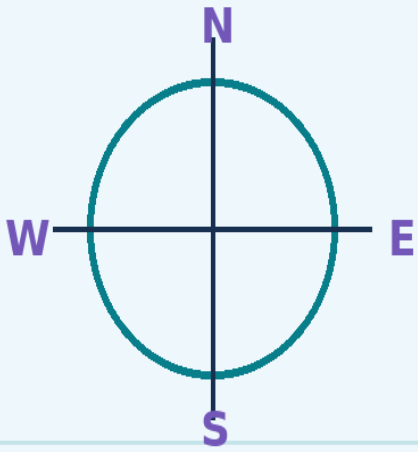
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Cardinal directions and routes

LEARNING GOAL Cardinal directions and routes

MATHS MODEL • DIRECTION



LEARN THE IDEA

The four cardinal directions are north, east, south and west; a quarter-turn moves to the next direction.

WORKED EXAMPLE 1

Face east. Turn a quarter-turn right. Which way now?

Answer: south

Use the compass order N–E–S–W; after turning, face south.

WORKED EXAMPLE 2

Face south. Turn a quarter-turn left. Which way now?

Answer: east

Use the compass order N–E–S–W; after turning, face east.

TRY THIS IN REAL LIFE

Use folded paper, a ruler or safe classroom objects to model each shape or direction.

Cardinal directions and routes

PLAY · LEARN THE STEPS

1. **Face east. Turn a quarter-turn left. Which way now?**
   A) north B) south C) east D) west
2. **Face east. Turn a quarter-turn right. Which way now?**
   A) north B) south C) east D) west
3. **Face south. Turn half a turn. Which way now?**
   A) south B) north C) west D) east
4. **Face west. Turn a quarter-turn right. Which way now?**
   A) north B) west C) east D) south
5. **Face south. Turn a quarter-turn left. Which way now?**
   A) south B) north C) west D) east

CHALLENGE · DIFFERENT EXAMPLES

1. **A learner faces east at a market stall and turns one quarter-turn to the right. Which direction is faced next?**
   A) north B) south C) east D) west
2. **A learner faces west during a library project and turns half a turn. Which direction is faced next?**
   A) north B) west C) south D) east
3. **A learner faces west on a football field and turns half a turn. Which direction is faced next?**
   A) north B) south C) west D) east
4. **A learner faces north in a recycling club and turns one quarter-turn to the left. Which direction is faced next?**
   A) west B) east C) south D) north
5. **A learner faces east at a science exhibition and turns one quarter-turn to the left. Which direction is faced next?**
   A) west B) east C) south D) north

Cardinal directions and routes

CHALLENGE · APPLY WHAT YOU KNOW

6 **A learner faces east during a reading competition and turns one quarter-turn to the right. Which direction is faced next?**

A) west B) south C) east D) north

7 **A learner faces east for a school fundraiser and turns half a turn. Which direction is faced next?**

A) west B) north C) south D) east

8 **A learner faces north at a farm and turns half a turn. Which direction is faced next?**

A) north B) south C) west D) east

9 **A learner faces south during a sports day and turns half a turn. Which direction is faced next?**

A) east B) west C) north D) south

10 **A learner faces east in a craft workshop and turns one quarter-turn to the left. Which direction is faced next?**

A) north B) east C) south D) west

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. north	2. south	3. north
4. north	5. east	6. south
7. east	8. east	9. west
10. north	11. south	12. west
13. south	14. north	15. north

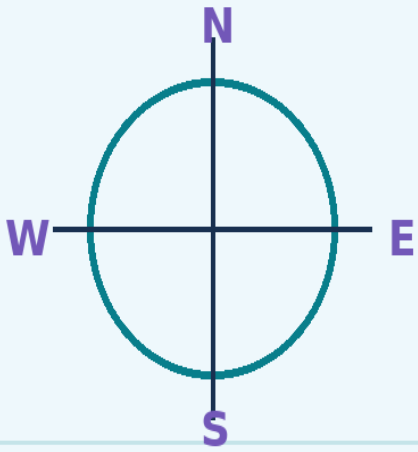
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Reflecting shapes across a mirror line

LEARNING GOAL Reflecting shapes across a mirror line

MATHS MODEL • REFLECTION



LEARN THE IDEA

A reflection is a mirror image across a line. every point stays the same distance from the line.

WORKED EXAMPLE 1

Reflect (4, 7) in the y-axis.

Answer: (-4, 7)

Reflection in the y-axis changes the sign of x only.

WORKED EXAMPLE 2

Reflect (6, 1) in the y-axis.

Answer: (-6, 1)

Reflection in the y-axis changes the sign of x only.

TRY THIS IN REAL LIFE

Use folded paper, a ruler or safe classroom objects to model each shape or direction.

Reflecting shapes across a mirror line

PLAY · LEARN THE STEPS

1. **Reflect (3, 3) in the y-axis.**
   A) (3, -3) B) (-3, -3) C) none of these D) (-3, 3)
2. **Reflect (7, 7) in the y-axis.**
   A) (-7, -7) B) none of these C) (7, -7) D) (-7, 7)
3. **Reflect (1, 3) in the y-axis.**
   A) (-1, 3) B) (1, -3) C) (-1, -3) D) (3, -1)
4. **Reflect (8, 7) in the y-axis.**
   A) (8, -7) B) (-8, 7) C) (-8, -7) D) (7, -8)
5. **Reflect (2, 3) in the y-axis.**
   A) (-2, 3) B) (3, -2) C) (2, -3) D) (-2, -3)

CHALLENGE · DIFFERENT EXAMPLES

1. **A point during a library project has coordinates (1, 7). Reflect it across the vertical y-axis. What is the new point?**
   A) (-1, 7) B) (1, -7) C) (7, -1) D) (-1, -7)
2. **A point on a football field has coordinates (8, 2). Reflect it across the vertical y-axis. What is the new point?**
   A) (2, -8) B) (-8, -2) C) (8, -2) D) (-8, 2)
3. **A point in a recycling club has coordinates (3, 7). Reflect it across the vertical y-axis. What is the new point?**
   A) (7, -3) B) (-3, -7) C) (3, -7) D) (-3, 7)
4. **A point at a science exhibition has coordinates (2, 3). Reflect it across the vertical y-axis. What is the new point?**
   A) (-2, 3) B) (2, -3) C) (3, -2) D) (-2, -3)
5. **A point during a reading competition has coordinates (8, 6). Reflect it across the vertical y-axis. What is the new point?**
   A) (-8, 6) B) (-8, -6) C) (6, -8) D) (8, -6)

Reflecting shapes across a mirror line

CHALLENGE · APPLY WHAT YOU KNOW

6 **A point for a school fundraiser has coordinates (1, 8). Reflect it across the vertical y-axis. What is the new point?**

A) (8, -1) B) (1, -8) C) (-1, -8) D) (-1, 8)

7 **A point at a farm has coordinates (4, 8). Reflect it across the vertical y-axis. What is the new point?**

A) (-4, 8) B) (-4, -8) C) (8, -4) D) (4, -8)

8 **A point during a sports day has coordinates (8, 6). Reflect it across the vertical y-axis. What is the new point?**

A) (8, -6) B) (-8, 6) C) (-8, -6) D) (6, -8)

9 **A point in a craft workshop has coordinates (7, 1). Reflect it across the vertical y-axis. What is the new point?**

A) (7, -1) B) (-7, -1) C) (1, -7) D) (-7, 1)

10 **A point at a school fair has coordinates (4, 3). Reflect it across the vertical y-axis. What is the new point?**

A) (3, -4) B) (4, -3) C) (-4, -3) D) (-4, 3)

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. (-3, 3)	2. (-7, 7)	3. (-1, 3)
4. (-8, 7)	5. (-2, 3)	6. (-1, 7)
7. (-8, 2)	8. (-3, 7)	9. (-2, 3)
10. (-8, 6)	11. (-1, 8)	12. (-4, 8)
13. (-8, 6)	14. (-7, 1)	15. (-4, 3)

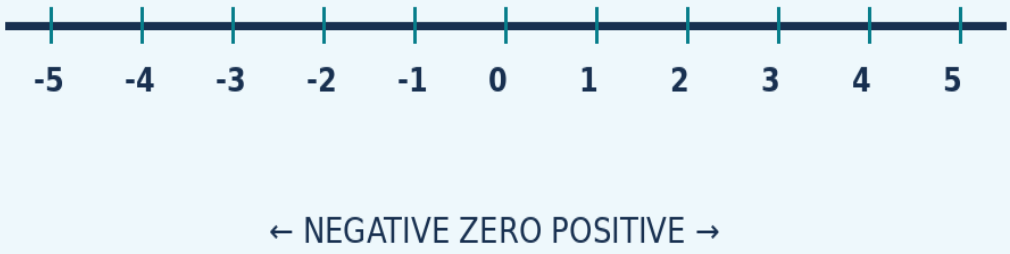
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Translating shapes across a grid

LEARNING GOAL Translating shapes across a grid

MATHS MODEL • TRANSLATION



LEARN THE IDEA

A translation slides every point the same distance and direction without turning or resizing the shape.

WORKED EXAMPLE 1

Translate point (-4, 1) by (3, 3). What is its new x-coordinate?

Answer: -1

Add +3 to the original x-coordinate -4: new x is -1.

WORKED EXAMPLE 2

Translate point (5, 5) by (-5, -2). What is its new x-coordinate?

Answer: 0

Add -5 to the original x-coordinate 5: new x is 0.

TRY THIS IN REAL LIFE

Use folded paper, a ruler or safe classroom objects to model each shape or direction.

Translating shapes across a grid

PLAY · LEARN THE STEPS

1. **Translate point (-4, -2) by (4, 3). What is its new x-coordinate?**
   A) -4 B) -8 C) 1 D) 0
2. **Translate point (5, 5) by (-5, -1). What is its new x-coordinate?**
   A) 1 B) 5 C) 0 D) 10
3. **Translate point (4, 1) by (5, -3). What is its new x-coordinate?**
   A) -1 B) 9 C) 10 D) 4
4. **Translate point (3, -2) by (-3, 4). What is its new x-coordinate?**
   A) 1 B) 6 C) 3 D) 0
5. **Translate point (-1, 0) by (-4, -2). What is its new x-coordinate?**
   A) -4 B) 3 C) -5 D) -1

CHALLENGE · DIFFERENT EXAMPLES

1. **A point during a reading competition moves from (-1, -2) by -1 squares horizontally and -2 vertically. What is its new x-coordinate?**
   A) 0 B) -2 C) -1 D) 3
2. **A point for a school fundraiser moves from (3, 3) by +1 squares horizontally and -3 vertically. What is its new x-coordinate?**
   A) 4 B) 2 C) 3 D) 5
3. **A point at a farm moves from (-2, 5) by -5 squares horizontally and +1 vertically. What is its new x-coordinate?**
   A) 3 B) -7 C) -6 D) -2
4. **A point during a sports day moves from (0, 1) by +2 squares horizontally and -4 vertically. What is its new x-coordinate?**
   A) -2 B) 0 C) 3 D) 2
5. **A point in a craft workshop moves from (-5, 1) by -1 squares horizontally and -2 vertically. What is its new x-coordinate?**
   A) -1 B) -4 C) -6 D) -5

Translating shapes across a grid

CHALLENGE · APPLY WHAT YOU KNOW

6 **A point at a school fair moves from (-3, -5) by +2 squares horizontally and +3 vertically. What is its new x-coordinate?**

A) -1 B) 0 C) -5 D) -3

7 **A point in the class garden moves from (-1, -3) by -4 squares horizontally and -2 vertically. What is its new x-coordinate?**

A) -4 B) -1 C) 3 D) -5

8 **A point at a market stall moves from (-1, 0) by +1 squares horizontally and -3 vertically. What is its new x-coordinate?**

A) -1 B) 1 C) 0 D) -2

9 **A point during a library project moves from (-2, -4) by -3 squares horizontally and +1 vertically. What is its new x-coordinate?**

A) -4 B) -2 C) -5 D) 1

10 **A point on a football field moves from (-1, -3) by +2 squares horizontally and -3 vertically. What is its new x-coordinate?**

A) 2 B) 1 C) -3 D) -1

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 0	2. 0	3. 9
4. 0	5. -5	6. -2
7. 4	8. -7	9. 2
10. -6	11. -1	12. -5
13. 0	14. -5	15. 1

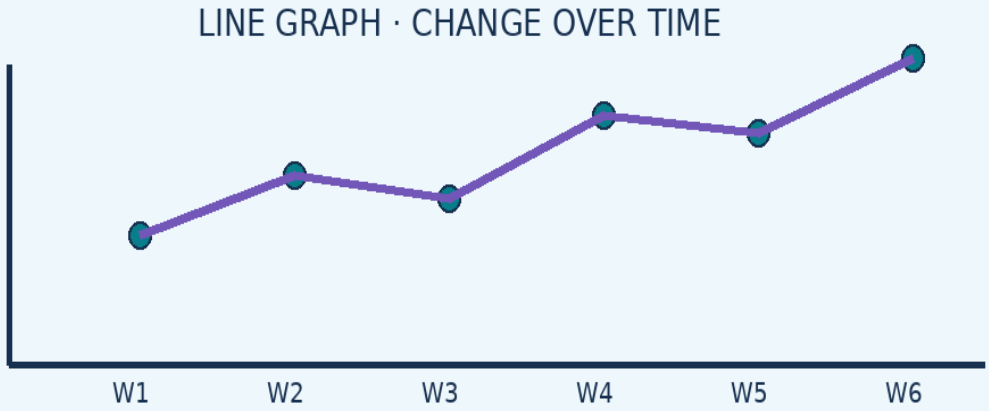
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Labelling and reading line graphs

LEARNING GOAL Labelling and reading line graphs

MATHS MODEL • GRAPH PARTS



LEARN THE IDEA

A meaningful line graph needs a title, labelled axes, sensible equal intervals and correctly plotted points.

WORKED EXAMPLE 1

What graph component is described as: the label along the horizontal axis?

Answer: x-axis label

This feature is called the x-axis label.

WORKED EXAMPLE 2

What graph component is described as: the measure on the vertical axis?

Answer: y-axis label

This feature is called the y-axis label.

TRY THIS IN REAL LIFE

Gather original class information, represent it in a table, and discuss what the chart reveals.

Labelling and reading line graphs

PLAY · LEARN THE STEPS

1. **What graph component is described as: the label along the horizontal axis?**
   A) title B) y-axis label C) scale D) x-axis label
2. **What graph component is described as: the measure on the vertical axis?**
   A) y-axis label B) x-axis label C) scale D) title
3. **What graph component is described as: a sentence describing the whole graph?**
   A) title B) x-axis label C) scale D) y-axis label
4. **What graph component is described as: evenly spaced number marks?**
   A) y-axis label B) x-axis label C) scale D) title
5. **What graph component is described as: colours identifying each drawn series?**
   A) y-axis label B) x-axis label C) title D) key

CHALLENGE · DIFFERENT EXAMPLES

1. **A rainfall line graph for a school fundraiser is missing: the measure on the vertical axis. Which feature should be added?**
   A) y-axis label B) scale C) x-axis label D) title
2. **A rainfall line graph at a farm is missing: a sentence describing the whole graph. Which feature should be added?**
   A) title B) x-axis label C) y-axis label D) scale
3. **A rainfall line graph during a sports day is missing: evenly spaced number marks. Which feature should be added?**
   A) scale B) x-axis label C) y-axis label D) title
4. **A rainfall line graph in a craft workshop is missing: colours identifying each drawn series. Which feature should be added?**
   A) x-axis label B) title C) y-axis label D) key
5. **A rainfall line graph at a school fair is missing: units of the measurements. Which feature should be added?**
   A) x-axis label B) units C) title D) y-axis label

Labelling and reading line graphs

CHALLENGE · APPLY WHAT YOU KNOW

6 **A rainfall line graph in the class garden is missing: the label along the horizontal axis. Which feature should be added?**

A) scale B) y-axis label C) title D) x-axis label

7 **A rainfall line graph at a market stall is missing: the measure on the vertical axis. Which feature should be added?**

A) y-axis label B) scale C) title D) x-axis label

8 **A rainfall line graph during a library project is missing: a sentence describing the whole graph. Which feature should be added?**

A) title B) scale C) x-axis label D) y-axis label

9 **A rainfall line graph on a football field is missing: evenly spaced number marks. Which feature should be added?**

A) title B) y-axis label C) x-axis label D) scale

10 **A rainfall line graph in a recycling club is missing: colours identifying each drawn series. Which feature should be added?**

A) x-axis label B) title C) key D) y-axis label

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. x-axis label	2. y-axis label	3. title
4. scale	5. key	6. y-axis label
7. title	8. scale	9. key
10. units	11. x-axis label	12. y-axis label
13. title	14. scale	15. key

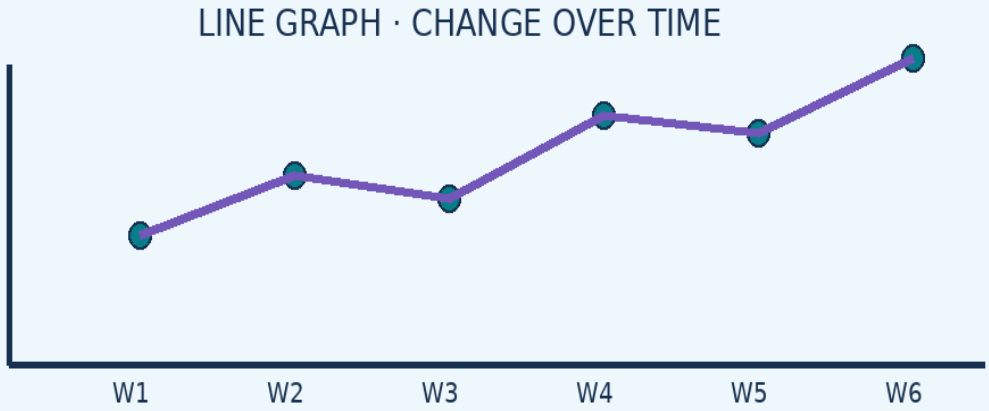
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Choosing when a line graph is suitable

LEARNING GOAL Choosing when a line graph is suitable

MATHS MODEL • GRAPH SUITABLE



LEARN THE IDEA

Line graphs are good for showing changes over ordered time; category comparisons often use bar graphs.

WORKED EXAMPLE 1

Choose a suitable graph for rainfall trend across seven successive days.

Answer: line graph

A line graph is suitable for rainfall trend across seven successive days.

WORKED EXAMPLE 2

Choose a suitable graph for number of learners per club.

Answer: bar graph

A bar graph is suitable for number of learners per club.

TRY THIS IN REAL LIFE

Gather original class information, represent it in a table, and discuss what the chart reveals.

Choosing when a line graph is suitable

PLAY · LEARN THE STEPS

1. **Choose a suitable graph for temperature changes hour by hour.**
   A) line graph B) bar graph C) pie chart D) pictograph
2. **Choose a suitable graph for favourite fruit categories.**
   A) bar graph B) line graph C) pictograph D) pie chart
3. **Choose a suitable graph for rainfall trend across seven successive days.**
   A) line graph B) pie chart C) pictograph D) bar graph
4. **Choose a suitable graph for number of learners per club.**
   A) pictograph B) pie chart C) bar graph D) line graph
5. **Choose a suitable graph for water level each minute.**
   A) pictograph B) pie chart C) bar graph D) line graph

CHALLENGE · DIFFERENT EXAMPLES

1. **A teacher at a farm collects data about favourite fruit categories. Which type of graph best communicates this?**
   A) pictograph B) pie chart C) bar graph D) line graph
2. **A teacher during a sports day collects data about rainfall trend across seven successive days. Which type of graph best communicates this?**
   A) bar graph B) pictograph C) line graph D) pie chart
3. **A teacher in a craft workshop collects data about number of learners per club. Which type of graph best communicates this?**
   A) pictograph B) line graph C) pie chart D) bar graph
4. **A teacher at a school fair collects data about water level each minute. Which type of graph best communicates this?**
   A) bar graph B) pictograph C) line graph D) pie chart
5. **A teacher in the class garden collects data about different types of fruit sold. Which type of graph best communicates this?**
   A) line graph B) bar graph C) pie chart D) pictograph

Choosing when a line graph is suitable

CHALLENGE · APPLY WHAT YOU KNOW

6 **A teacher at a market stall collects data about trend in daily electricity use over a week. Which type of graph best communicates this?**

A) bar graph B) pictograph C) line graph D) pie chart

7 **A teacher during a library project collects data about eye colours of students. Which type of graph best communicates this?**

A) pictograph B) bar graph C) line graph D) pie chart

8 **A teacher on a football field collects data about temperature changes hour by hour. Which type of graph best communicates this?**

A) pictograph B) pie chart C) bar graph D) line graph

9 **A teacher in a recycling club collects data about favourite fruit categories. Which type of graph best communicates this?**

A) pie chart B) line graph C) bar graph D) pictograph

10 **A teacher at a science exhibition collects data about rainfall trend across seven successive days. Which type of graph best communicates this?**

A) pie chart B) line graph C) bar graph D) pictograph

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. line graph	2. bar graph	3. line graph
4. bar graph	5. line graph	6. bar graph
7. line graph	8. bar graph	9. line graph
10. bar graph	11. line graph	12. bar graph
13. line graph	14. bar graph	15. line graph

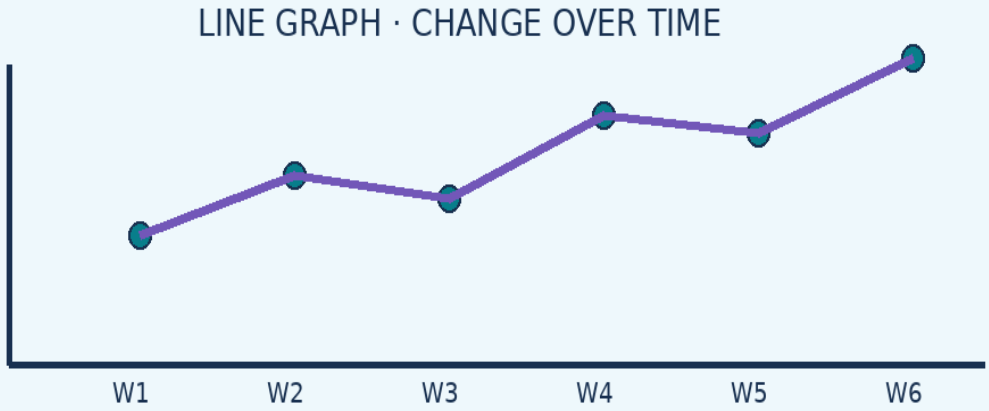
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Plotting and interpreting line graphs

LEARNING GOAL Plotting and interpreting line graphs

MATHS MODEL • LINE GRAPH



LEARN THE IDEA

Use equally spaced axes and plot coordinate pairs (time, measurement), then connect in order.

WORKED EXAMPLE 1

A line graph begins at 13 on day 1 and increases by 4 daily. Find day 5.

Answer: 29

Value on day 5 is $13+(5-1)\times4=29$.

WORKED EXAMPLE 2

A line graph begins at 19 on day 1 and increases by 6 daily. Find day 2.

Answer: 25

Value on day 2 is $19+(2-1)\times6=25$.

TRY THIS IN REAL LIFE

Gather original class information, represent it in a table, and discuss what the chart reveals.

Plotting and interpreting line graphs

PLAY · LEARN THE STEPS

1. **A line graph begins at 12 on day 1 and increases by 6 daily. Find day 5.**
   A) 37 B) 36 C) 30 D) 42
2. **A line graph begins at 18 on day 1 and increases by 5 daily. Find day 2.**
   A) 23 B) 24 C) 18 D) 28
3. **A line graph begins at 8 on day 1 and increases by 3 daily. Find day 4.**
   A) 14 B) 18 C) 20 D) 17
4. **A line graph begins at 17 on day 1 and increases by 2 daily. Find day 3.**
   A) 22 B) 21 C) 19 D) 23
5. **A line graph begins at 19 on day 1 and increases by 5 daily. Find day 5.**
   A) 44 B) 39 C) 34 D) 40

CHALLENGE · DIFFERENT EXAMPLES

1. **The reading chart during a sports day shows 12 books on day 1 and rises by 5 each day. What y-value is plotted for day 2?**
   A) 12 B) 18 C) 22 D) 17
2. **The reading chart in a craft workshop shows 21 books on day 1 and rises by 6 each day. What y-value is plotted for day 5?**
   A) 51 B) 45 C) 39 D) 46
3. **The reading chart at a school fair shows 14 books on day 1 and rises by 6 each day. What y-value is plotted for day 4?**
   A) 33 B) 26 C) 32 D) 38
4. **The reading chart in the class garden shows 16 books on day 1 and rises by 6 each day. What y-value is plotted for day 4?**
   A) 35 B) 34 C) 28 D) 40
5. **The reading chart at a market stall shows 13 books on day 1 and rises by 6 each day. What y-value is plotted for day 5?**
   A) 38 B) 43 C) 31 D) 37

Plotting and interpreting line graphs

CHALLENGE · APPLY WHAT YOU KNOW

6 **The reading chart during a library project shows 22 books on day 1 and rises by 4 each day. What y-value is plotted for day 4?**

A) 30 B) 38 C) 34 D) 35

7 **The reading chart on a football field shows 10 books on day 1 and rises by 5 each day. What y-value is plotted for day 4?**

A) 20 B) 30 C) 25 D) 26

8 **The reading chart in a recycling club shows 12 books on day 1 and rises by 4 each day. What y-value is plotted for day 3?**

A) 21 B) 24 C) 20 D) 16

9 **The reading chart at a science exhibition shows 14 books on day 1 and rises by 5 each day. What y-value is plotted for day 4?**

A) 24 B) 30 C) 34 D) 29

10 **The reading chart during a reading competition shows 11 books on day 1 and rises by 3 each day. What y-value is plotted for day 2?**

A) 15 B) 14 C) 11 D) 17

ANSWER CHECK · LOOK ONLY AFTER TRYING

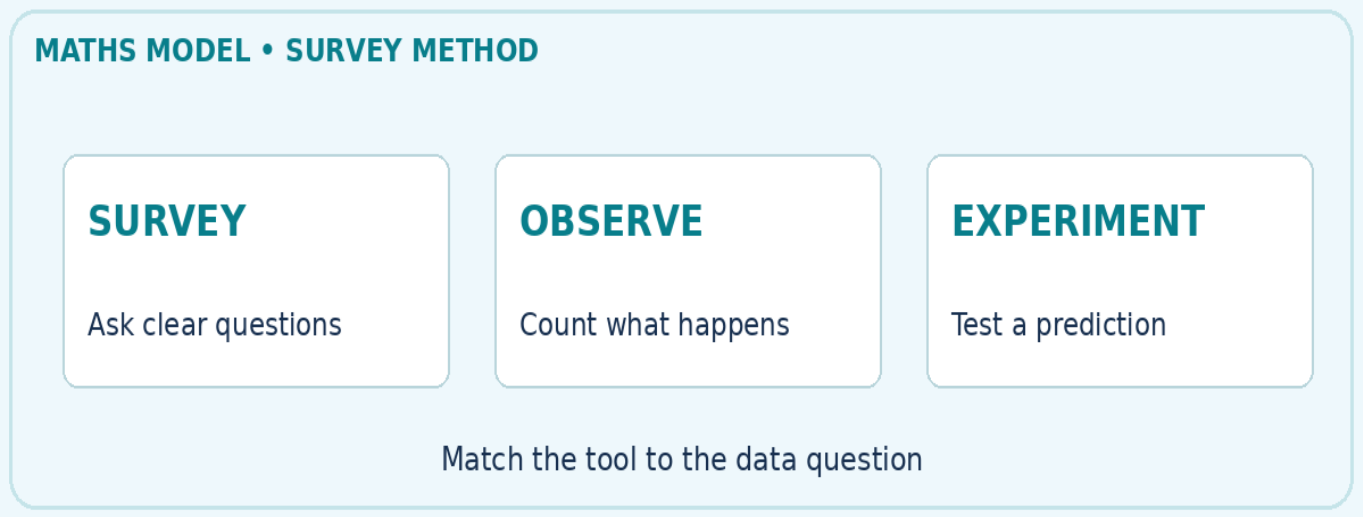
1. 36	2. 23	3. 17
4. 21	5. 39	6. 17
7. 45	8. 32	9. 34
10. 37	11. 34	12. 25
13. 20	14. 29	15. 14

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Choosing how to collect data

LEARNING GOAL Choosing how to collect data



LEARN THE IDEA

Choose data collection tools to suit the question: surveys, observations, experiments or reliable records.

WORKED EXAMPLE 1

Choose the data collection method for how fast seeds grow.

Answer: experiment

The most direct reliable method here is experiment.

WORKED EXAMPLE 2

Choose the data collection method for national census from an official report.

Answer: secondary records

The most direct reliable method here is secondary records.

TRY THIS IN REAL LIFE

Gather original class information, represent it in a table, and discuss what the chart reveals.

Choosing how to collect data

PLAY · LEARN THE STEPS

1. **Choose the data collection method for favourite school subject.**
   A) questionnaire B) secondary records C) observation D) experiment
2. **Choose the data collection method for number of cars passing a gate.**
   A) observation B) questionnaire C) experiment D) secondary records
3. **Choose the data collection method for how fast seeds grow.**
   A) secondary records B) experiment C) questionnaire D) observation
4. **Choose the data collection method for national census from an official report.**
   A) observation B) experiment C) secondary records D) questionnaire
5. **Choose the data collection method for how many pupils like each fruit.**
   A) observation B) experiment C) questionnaire D) secondary records

CHALLENGE · DIFFERENT EXAMPLES

1. **A class in a craft workshop wants to find the number of cars passing a gate. Which data collection method should they use?**
   A) questionnaire B) experiment C) observation D) secondary records
2. **A class at a school fair wants to find the how fast seeds grow. Which data collection method should they use?**
   A) questionnaire B) experiment C) observation D) secondary records
3. **A class in the class garden wants to find the national census from an official report. Which data collection method should they use?**
   A) questionnaire B) experiment C) observation D) secondary records
4. **A class at a market stall wants to find the how many pupils like each fruit. Which data collection method should they use?**
   A) questionnaire B) observation C) secondary records D) experiment
5. **A class during a library project wants to find the number of birds in a tree. Which data collection method should they use?**
   A) questionnaire B) observation C) secondary records D) experiment

Choosing how to collect data

CHALLENGE · APPLY WHAT YOU KNOW

6 **A class on a football field wants to find the how light affects plants. Which data collection method should they use?**

A) questionnaire B) experiment C) observation D) secondary records

7 **A class in a recycling club wants to find the published rainfall data for Ghana. Which data collection method should they use?**

A) observation B) questionnaire C) secondary records D) experiment

8 **A class at a science exhibition wants to find the favourite school subject. Which data collection method should they use?**

A) questionnaire B) observation C) experiment D) secondary records

9 **A class during a reading competition wants to find the number of cars passing a gate. Which data collection method should they use?**

A) secondary records B) experiment C) questionnaire D) observation

10 **A class for a school fundraiser wants to find the how fast seeds grow. Which data collection method should they use?**

A) questionnaire B) secondary records C) observation D) experiment

ANSWER CHECK · LOOK ONLY AFTER TRYING

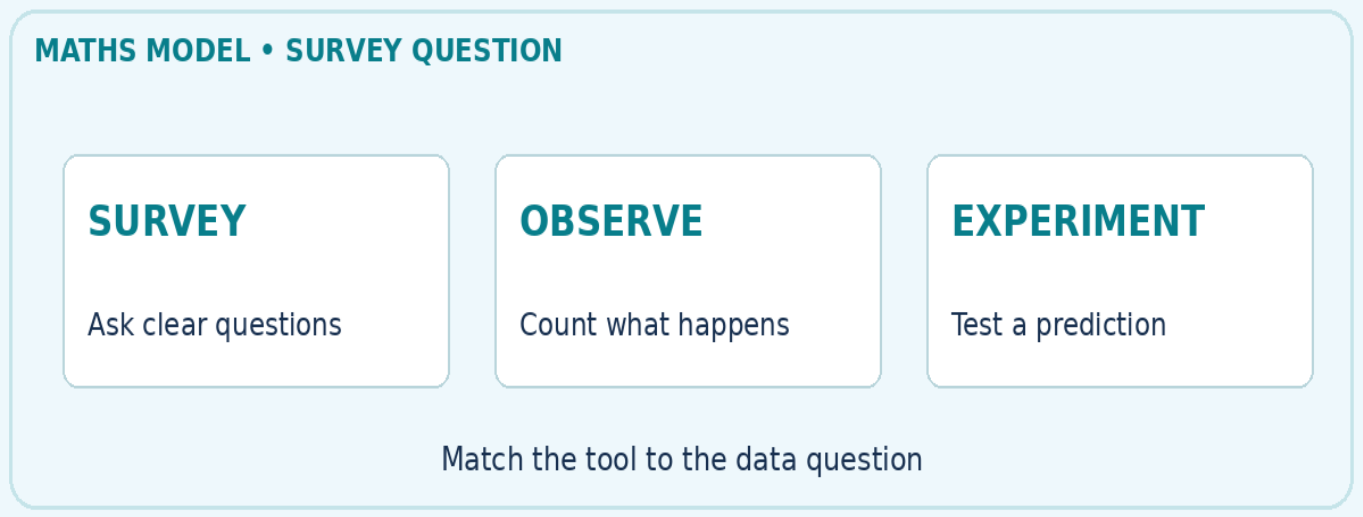
1. questionnaire	2. observation	3. experiment
4. secondary records	5. questionnaire	6. observation
7. experiment	8. secondary records	9. questionnaire
10. observation	11. experiment	12. secondary records
13. questionnaire	14. observation	15. experiment

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Designing useful survey questions

LEARNING GOAL Designing useful survey questions



LEARN THE IDEA

A good questionnaire uses neutral, precise and single-purpose questions with suitable answer options.

WORKED EXAMPLE 1

Classify this survey question: “Is it good?”

Answer: unclear

Is it good? is classified as unclear; surveys work best with precise neutral wording.

WORKED EXAMPLE 2

Classify this survey question: “Did you walk AND cycle to school today?”

Answer: double question

Did you walk AND cycle to school today? is classified as double question; surveys work best with precise neutral wording.

TRY THIS IN REAL LIFE

Gather original class information, represent it in a table, and discuss what the chart reveals.

Designing useful survey questions

PLAY · LEARN THE STEPS

1. **Classify this survey question: “Which fruit do you like best?”**
   A) clear and neutral B) unclear C) double question D) leading
2. **Classify this survey question: “Do you agree that football is the only fun sport?”**
   A) unclear B) double question C) clear and neutral D) leading
3. **Classify this survey question: “Is it good?”**
   A) clear and neutral B) leading C) double question D) unclear
4. **Classify this survey question: “Did you walk AND cycle to school today?”**
   A) double question B) unclear C) leading D) clear and neutral
5. **Classify this survey question: “What time do you arrive at school?”**
   A) double question B) leading C) clear and neutral D) unclear

CHALLENGE · DIFFERENT EXAMPLES

1. **A student at a school fair asks “Do you agree that football is the only fun sport?” in a survey. What describes this question?**
   A) unclear B) double question C) leading D) clear and neutral
2. **A student in the class garden asks “Is it good?” in a survey. What describes this question?**
   A) leading B) unclear C) clear and neutral D) double question
3. **A student at a market stall asks “Did you walk AND cycle to school today?” in a survey. What describes this question?**
   A) unclear B) double question C) leading D) clear and neutral
4. **A student during a library project asks “What time do you arrive at school?” in a survey. What describes this question?**
   A) double question B) clear and neutral C) leading D) unclear
5. **A student on a football field asks “Everyone loves maths, do you?” in a survey. What describes this question?**
   A) unclear B) double question C) leading D) clear and neutral

Designing useful survey questions

CHALLENGE · APPLY WHAT YOU KNOW

6 **A student in a recycling club asks “Is your family big?” in a survey. What describes this question?**

A) clear and neutral B) double question C) leading D) unclear

7 **A student at a science exhibition asks “Do you enjoy art AND physical education?” in a survey. What describes this question?**

A) leading B) double question C) unclear D) clear and neutral

8 **A student during a reading competition asks “Which fruit do you like best?” in a survey. What describes this question?**

A) double question B) unclear C) clear and neutral D) leading

9 **A student for a school fundraiser asks “Do you agree that football is the only fun sport?” in a survey. What describes this question?**

A) unclear B) leading C) clear and neutral D) double question

10 **A student at a farm asks “Is it good?” in a survey. What describes this question?**

A) double question B) clear and neutral C) unclear D) leading

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. clear and neutral	2. leading	3. unclear
4. double question	5. clear and neutral	6. leading
7. unclear	8. double question	9. clear and neutral
10. leading	11. unclear	12. double question
13. clear and neutral	14. leading	15. unclear

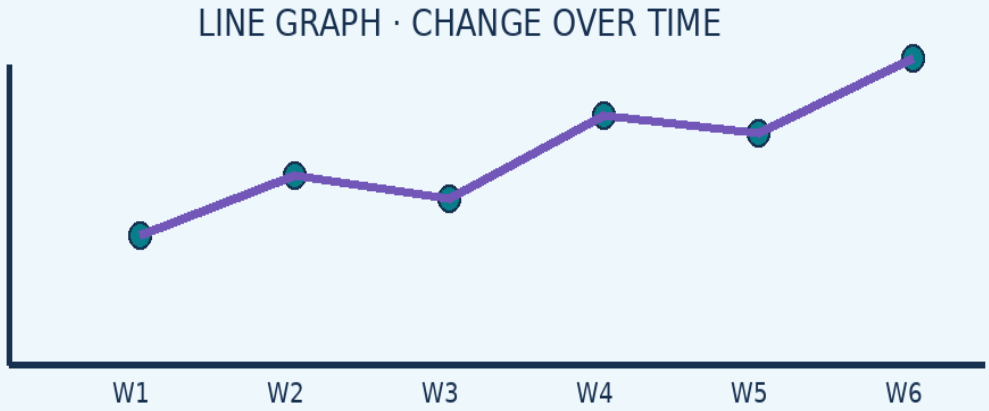
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Organising survey results

LEARNING GOAL Organising survey results

MATHS MODEL • SURVEY TABLE



LEARN THE IDEA

A frequency table counts observations in categories. add the category frequencies to verify the total.

WORKED EXAMPLE 1

Which is the highest frequency in [7, 7, 3, 4]?

Answer: 7

The maximum entry in the frequency table is 7.

WORKED EXAMPLE 2

Find the total frequency of [4, 7, 5, 4].

Answer: 20

Add all four values to obtain 20.

TRY THIS IN REAL LIFE

Gather original class information, represent it in a table, and discuss what the chart reveals.

Organising survey results

PLAY · LEARN THE STEPS

1. **Which is the highest frequency in [3, 5, 3, 4]?**
   A) 14 B) 5 C) 3 D) 15
2. **Find the total frequency of [4, 3, 4, 3].**
   A) 16 B) 4 C) 14 D) 12
3. **Which is the highest frequency in [8, 4, 5, 5]?**
   A) 4 B) 22 C) 8 D) 21
4. **Find the total frequency of [10, 7, 4, 10].**
   A) 31 B) 29 C) 10 D) 33
5. **Which is the highest frequency in [8, 9, 7, 6]?**
   A) 6 B) 30 C) 29 D) 9

CHALLENGE · DIFFERENT EXAMPLES

1. **A survey in the class garden records four groups with frequencies [10, 6, 5, 4]. Which is the total frequency?**
   A) 23 B) 10 C) 25 D) 27
2. **The groups at a market stall have frequencies [7, 8, 5, 4]. What is the largest frequency?**
   A) 8 B) 24 C) 23 D) 4
3. **A survey during a library project records four groups with frequencies [10, 8, 4, 8]. Which is the total frequency?**
   A) 32 B) 30 C) 10 D) 28
4. **The groups on a football field have frequencies [7, 9, 8, 4]. What is the largest frequency?**
   A) 27 B) 9 C) 4 D) 28
5. **A survey in a recycling club records four groups with frequencies [7, 9, 9, 6]. Which is the total frequency?**
   A) 29 B) 9 C) 31 D) 33

Organising survey results

CHALLENGE · APPLY WHAT YOU KNOW

6 **The groups at a science exhibition have frequencies [9, 6, 8, 5]. What is the largest frequency?**

A) 9 B) 27 C) 28 D) 5

7 **A survey during a reading competition records four groups with frequencies [7, 9, 5, 7]. Which is the total frequency?**

A) 26 B) 9 C) 30 D) 28

8 **The groups for a school fundraiser have frequencies [9, 10, 10, 10]. What is the largest frequency?**

A) 9 B) 39 C) 38 D) 10

9 **A survey at a farm records four groups with frequencies [8, 5, 8, 8]. Which is the total frequency?**

A) 31 B) 27 C) 8 D) 29

10 **The groups during a sports day have frequencies [8, 5, 4, 10]. What is the largest frequency?**

A) 10 B) 27 C) 4 D) 26

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 5	2. 14	3. 8
4. 31	5. 9	6. 25
7. 8	8. 30	9. 9
10. 31	11. 9	12. 28
13. 10	14. 29	15. 10

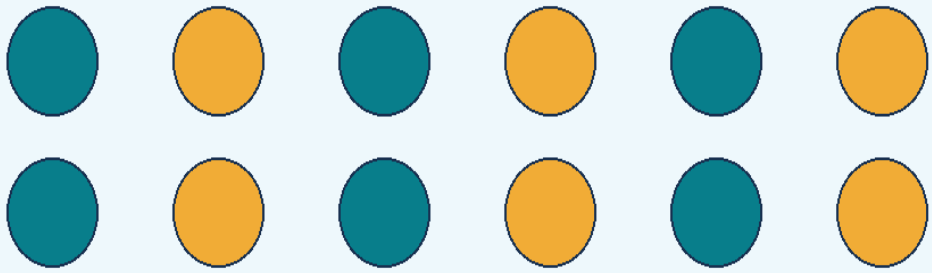
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Listing outcomes of chance experiments

LEARNING GOAL Listing outcomes of chance experiments

MATHS MODEL • PROB OUTCOMES



Collect results and compare probability

LEARN THE IDEA

A sample space is a complete list of all possible results of a fair chance experiment.

WORKED EXAMPLE 1

List all possible outcomes for a spinner labelled red, blue, green.

Answer: red, blue, green

The sample space for a spinner labelled red, blue, green is red, blue, green.

WORKED EXAMPLE 2

List all possible outcomes for a card numbered 1 through 4.

Answer: 1, 2, 3, 4

The sample space for a card numbered 1 through 4 is 1, 2, 3, 4.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Listing outcomes of chance experiments

PLAY · LEARN THE STEPS

1. **List all possible outcomes for a fair coin.**
   A) 1, 3, 5 B) heads, heads C) red, blue D) heads, tails
2. **List all possible outcomes for a numbered six-sided die.**
   A) heads, heads B) 1, 3, 5 C) red, blue D) 1, 2, 3, 4, 5, 6
3. **List all possible outcomes for a spinner labelled red, blue, green.**
   A) red, blue, green B) 1, 3, 5 C) heads, heads D) red, blue
4. **List all possible outcomes for a card numbered 1 through 4.**
   A) 1, 3, 5 B) heads, heads C) red, blue D) 1, 2, 3, 4
5. **List all possible outcomes for a two-colour bead draw.**
   A) black, white B) red, blue C) 1, 3, 5 D) heads, heads

CHALLENGE · DIFFERENT EXAMPLES

1. **A class at a market stall experiments with a numbered six-sided die. Which list has all possible outcomes?**
   A) heads, heads B) 1, 2, 3, 4, 5, 6 C) red, blue D) 1, 3, 5
2. **A class during a library project experiments with a spinner labelled red, blue, green. Which list has all possible outcomes?**
   A) red, blue, green B) red, blue C) heads, heads D) 1, 3, 5
3. **A class on a football field experiments with a card numbered 1 through 4. Which list has all possible outcomes?**
   A) 1, 2, 3, 4 B) 1, 3, 5 C) heads, heads D) red, blue
4. **A class in a recycling club experiments with a two-colour bead draw. Which list has all possible outcomes?**
   A) 1, 3, 5 B) heads, heads C) red, blue D) black, white
5. **A class at a science exhibition experiments with a spinner labelled A, B, C, D. Which list has all possible outcomes?**
   A) red, blue B) A, B, C, D C) 1, 3, 5 D) heads, heads

Listing outcomes of chance experiments

CHALLENGE · APPLY WHAT YOU KNOW

6 **A class during a reading competition experiments with a card marked yes or no. Which list has all possible outcomes?**

A) heads, heads B) 1, 3, 5 C) red, blue D) yes, no

7 **A class for a school fundraiser experiments with a die numbered 1 through 3 twice. Which list has all possible outcomes?**

A) red, blue B) heads, heads C) 1, 2, 3 D) 1, 3, 5

8 **A class at a farm experiments with a fair coin. Which list has all possible outcomes?**

A) red, blue B) heads, heads C) 1, 3, 5 D) heads, tails

9 **A class during a sports day experiments with a numbered six-sided die. Which list has all possible outcomes?**

A) red, blue B) 1, 3, 5 C) 1, 2, 3, 4, 5, 6 D) heads, heads

10 **A class in a craft workshop experiments with a spinner labelled red, blue, green. Which list has all possible outcomes?**

A) red, blue B) red, blue, green C) 1, 3, 5 D) heads, heads

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. heads, tails
2. 1, 2, 3, 4, 5, 6
3. red, blue, green
4. 1, 2, 3, 4
5. black, white
6. 1, 2, 3, 4, 5, 6
7. red, blue, green
8. 1, 2, 3, 4
9. black, white
10. A, B, C, D
11. yes, no
12. 1, 2, 3
13. heads, tails
14. 1, 2, 3, 4, 5, 6
15. red, blue, green

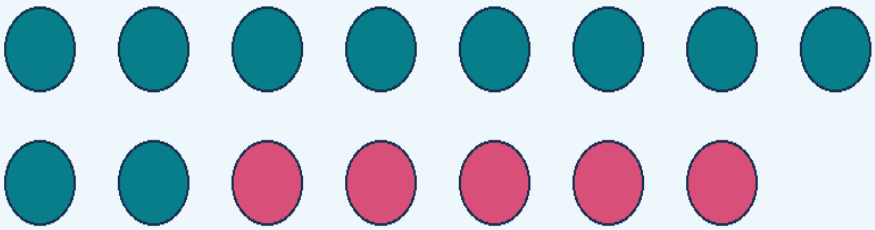
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Predicting chances and likelihood

LEARNING GOAL Predicting chances and likelihood

MATHS MODEL • PROBABILITY



Predict • Draw • Tally • Compare

LEARN THE IDEA

Predict and test an event using repeated fair trials. record all results and compare the observed frequencies.

WORKED EXAMPLE 1

Which is more likely: red (6) or blue (2) bottle tops?

Answer: red

Compare the counts; 6 red versus 2 blue.

WORKED EXAMPLE 2

Which is more likely: red (6) or blue (1) bottle tops?

Answer: red

Compare the counts; 6 red versus 1 blue.

TRY THIS IN REAL LIFE

Place safe coloured bottle tops in a bag, predict outcomes, then draw and tally twenty times.

Predicting chances and likelihood

PLAY · LEARN THE STEPS

1. **Which is more likely: red (4) or blue (6) bottle tops?**
   A) red B) equally likely C) neither can be picked D) blue
2. **Which is more likely: red (8) or blue (8) bottle tops?**
   A) neither can be picked B) red C) blue D) equally likely
3. **Which is more likely: red (9) or blue (7) bottle tops?**
   A) blue B) equally likely C) red D) neither can be picked
4. **Which is more likely: red (7) or blue (8) bottle tops?**
   A) neither can be picked B) red C) blue D) equally likely
5. **Which is more likely: red (4) or blue (9) bottle tops?**
   A) blue B) red C) equally likely D) neither can be picked

CHALLENGE · DIFFERENT EXAMPLES

1. **A bag at a school fair has 6 red and 5 blue bottle tops. Which colour is more likely on one random draw?**
   A) blue B) red C) equally likely D) neither can be picked
2. **A bag in the class garden has 8 red and 2 blue bottle tops. Which colour is more likely on one random draw?**
   A) blue B) equally likely C) red D) neither can be picked
3. **A bag at a market stall has 2 red and 1 blue bottle tops. Which colour is more likely on one random draw?**
   A) red B) blue C) neither can be picked D) equally likely
4. **A bag during a library project has 1 red and 4 blue bottle tops. Which colour is more likely on one random draw?**
   A) red B) blue C) equally likely D) neither can be picked
5. **A bag on a football field has 10 red and 3 blue bottle tops. Which colour is more likely on one random draw?**
   A) equally likely B) blue C) red D) neither can be picked

Predicting chances and likelihood

CHALLENGE · APPLY WHAT YOU KNOW

6 **A bag in a recycling club has 5 red and 6 blue bottle tops. Which colour is more likely on one random draw?**

A) red B) blue C) equally likely D) neither can be picked

7 **A bag at a science exhibition has 5 red and 5 blue bottle tops. Which colour is more likely on one random draw?**

A) equally likely B) blue C) red D) neither can be picked

8 **A bag during a reading competition has 9 red and 4 blue bottle tops. Which colour is more likely on one random draw?**

A) red B) neither can be picked C) blue D) equally likely

9 **A bag for a school fundraiser has 2 red and 6 blue bottle tops. Which colour is more likely on one random draw?**

A) blue B) red C) equally likely D) neither can be picked

10 **A bag at a farm has 1 red and 7 blue bottle tops. Which colour is more likely on one random draw?**

A) equally likely B) blue C) neither can be picked D) red

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. blue	2. equally likely	3. red
4. blue	5. blue	6. red
7. red	8. red	9. blue
10. red	11. blue	12. equally likely
13. red	14. blue	15. blue

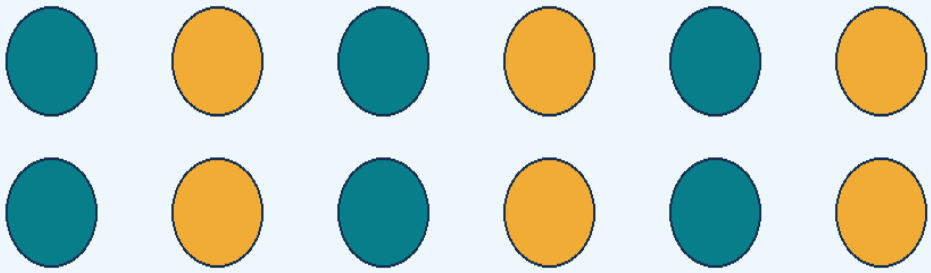
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Comparing experimental and theoretical probability

LEARNING GOAL Comparing experimental and theoretical probability

MATHS MODEL • EXP PROBABILITY



Collect results and compare probability

LEARN THE IDEA

Observed frequency divided by number of trials is experimental probability; it approaches theory with more trials.

WORKED EXAMPLE 1

Write the observed heads frequency as a fraction: 45 heads in 90 trials.

Answer: 45/90

Experimental probability = observed heads ÷ total tosses = 45/90; it may differ from 1/2.

WORKED EXAMPLE 2

Write the observed heads frequency as a fraction: 28 heads in 70 trials.

Answer: 28/70

Experimental probability = observed heads ÷ total tosses = 28/70; it may differ from 1/2.

TRY THIS IN REAL LIFE

Work with number cards, bottle tops or a number line. Predict, model, solve and explain to a partner.

Comparing experimental and theoretical probability

PLAY · LEARN THE STEPS

1. **Write the observed heads frequency as a fraction: 59 heads in 120 trials.**
   A) 61/120 B) 59/130 C) 120/59 D) 59/120
2. **Write the observed heads frequency as a fraction: 66 heads in 120 trials.**
   A) 66/130 B) 120/66 C) 66/120 D) 54/120
3. **Write the observed heads frequency as a fraction: 84 heads in 160 trials.**
   A) 160/84 B) 76/160 C) 84/160 D) 84/170
4. **Write the observed heads frequency as a fraction: 79 heads in 150 trials.**
   A) 79/150 B) 79/160 C) 150/79 D) 71/150
5. **Write the observed heads frequency as a fraction: 52 heads in 90 trials.**
   A) 90/52 B) 38/90 C) 52/90 D) 52/100

CHALLENGE · DIFFERENT EXAMPLES

1. **In a probability experiment on a football field, a coin landed heads 32 times in 50 tosses. What is the experimental probability before simplification?**
   A) 18/50 B) 32/60 C) 50/32 D) 32/50
2. **In a probability experiment in a recycling club, a coin landed heads 79 times in 160 tosses. What is the experimental probability before simplification?**
   A) 160/79 B) 79/170 C) 81/160 D) 79/160
3. **In a probability experiment at a science exhibition, a coin landed heads 32 times in 80 tosses. What is the experimental probability before simplification?**
   A) 80/32 B) 32/90 C) 48/80 D) 32/80
4. **In a probability experiment during a reading competition, a coin landed heads 72 times in 140 tosses. What is the experimental probability before simplification?**
   A) 72/140 B) 68/140 C) 72/150 D) 140/72
5. **In a probability experiment for a school fundraiser, a coin landed heads 70 times in 140 tosses. What is the experimental probability before simplification?**
   A) 70/140 B) 71/140 C) 70/150 D) 140/70

Comparing experimental and theoretical probability

CHALLENGE · APPLY WHAT YOU KNOW

6 **In a probability experiment at a farm, a coin landed heads 36 times in 80 tosses. What is the experimental probability before simplification?**

A) 80/36 B) 36/90 C) 44/80 D) 36/80

7 **In a probability experiment during a sports day, a coin landed heads 51 times in 100 tosses. What is the experimental probability before simplification?**

A) 49/100 B) 100/51 C) 51/100 D) 51/110

8 **In a probability experiment in a craft workshop, a coin landed heads 52 times in 100 tosses. What is the experimental probability before simplification?**

A) 52/110 B) 52/100 C) 48/100 D) 100/52

9 **In a probability experiment at a school fair, a coin landed heads 60 times in 110 tosses. What is the experimental probability before simplification?**

A) 60/120 B) 110/60 C) 50/110 D) 60/110

10 **In a probability experiment in the class garden, a coin landed heads 34 times in 80 tosses. What is the experimental probability before simplification?**

A) 80/34 B) 34/80 C) 46/80 D) 34/90

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 59/120	2. 66/120	3. 84/160
4. 79/150	5. 52/90	6. 32/50
7. 79/160	8. 32/80	9. 72/140
10. 70/140	11. 36/80	12. 51/100
13. 52/100	14. 60/110	15. 34/80

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.